
ELEC6095 Smart Grid

Electricity Pricing and Demand Response

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Demand Response / Demand Side Management



Background

Today power grid



Generation must follow demand to ensure power balance at all time



No dynamic pricing (i.e. fixed tariff)



Limited information in demand sides



Unable to accommodate large-scale unpredictable renewable generation

Future grid

- ▶ Demand is more actively engaged with power grid
- ▶ Cheap energy prices when the wind blows
- ▶ Integration of renewable generation becomes easier and more cost-effective

Drivers

Changing Supply

- Large penetration of unpredictable renewable energy resources
- Distributed and micro generation source integration

Changing Demand

- Consumer demand for renewable energy
- Consumers are more actively engaged in grid
- Flexible pricing to shift demand

Regulation

- Climate change and energy efficiency objectives
- Standards and interoperability
- Mandated smart meter rollouts

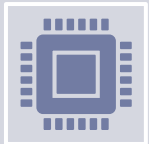
New Opportunities

- Consumer participation in delivery chain
- New service and business models
- Introduction of feed-in tariff

Demand Side Management (DSM)

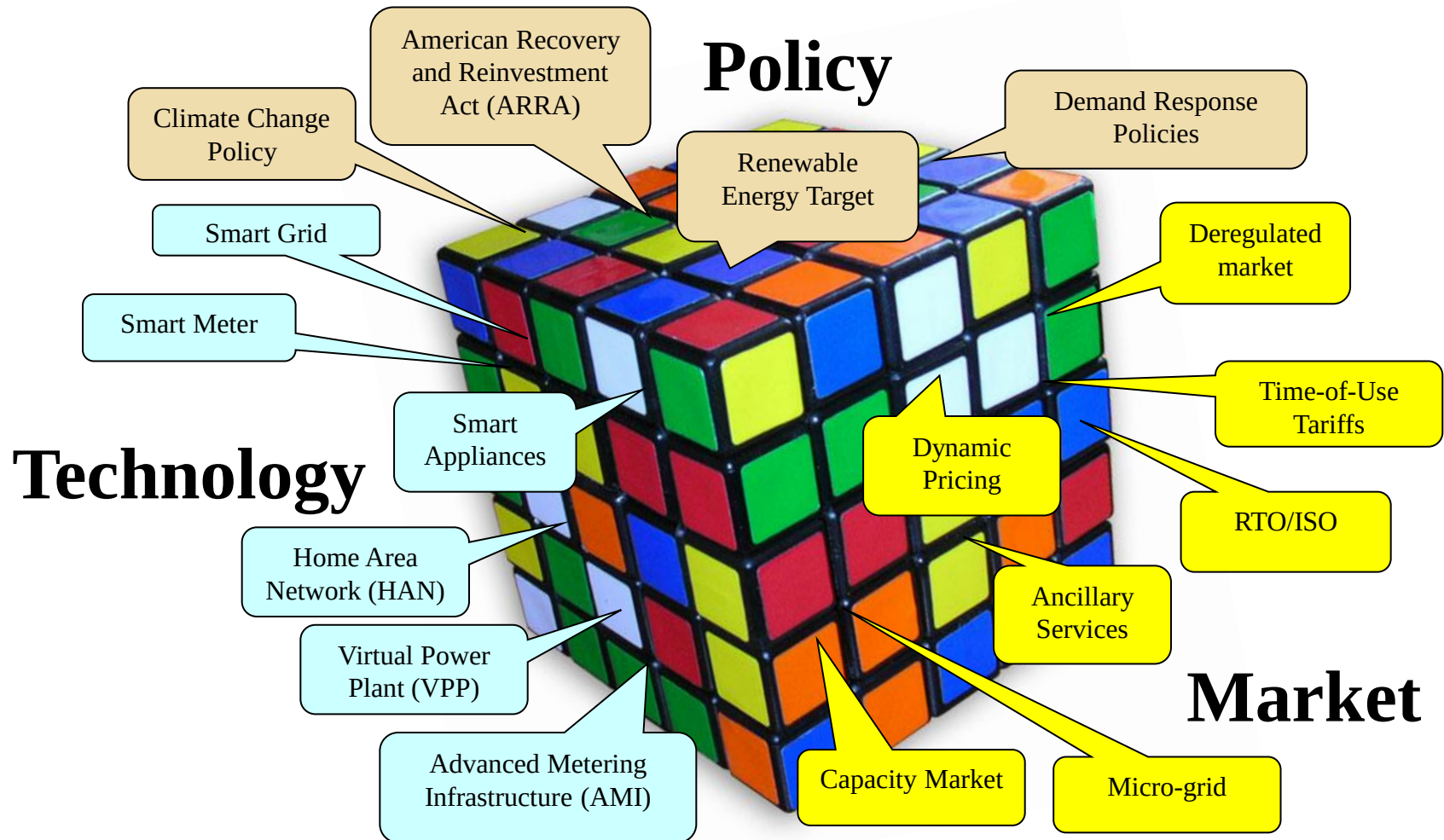


The objective is to produce desired changes in the utility's load shape (pattern and magnitude)



DSM can be found in all customer classes including residential, commercial, and industrial customers

Precursors/Components of Demand Side Management



Definition of Demand Response/Demand Side Management

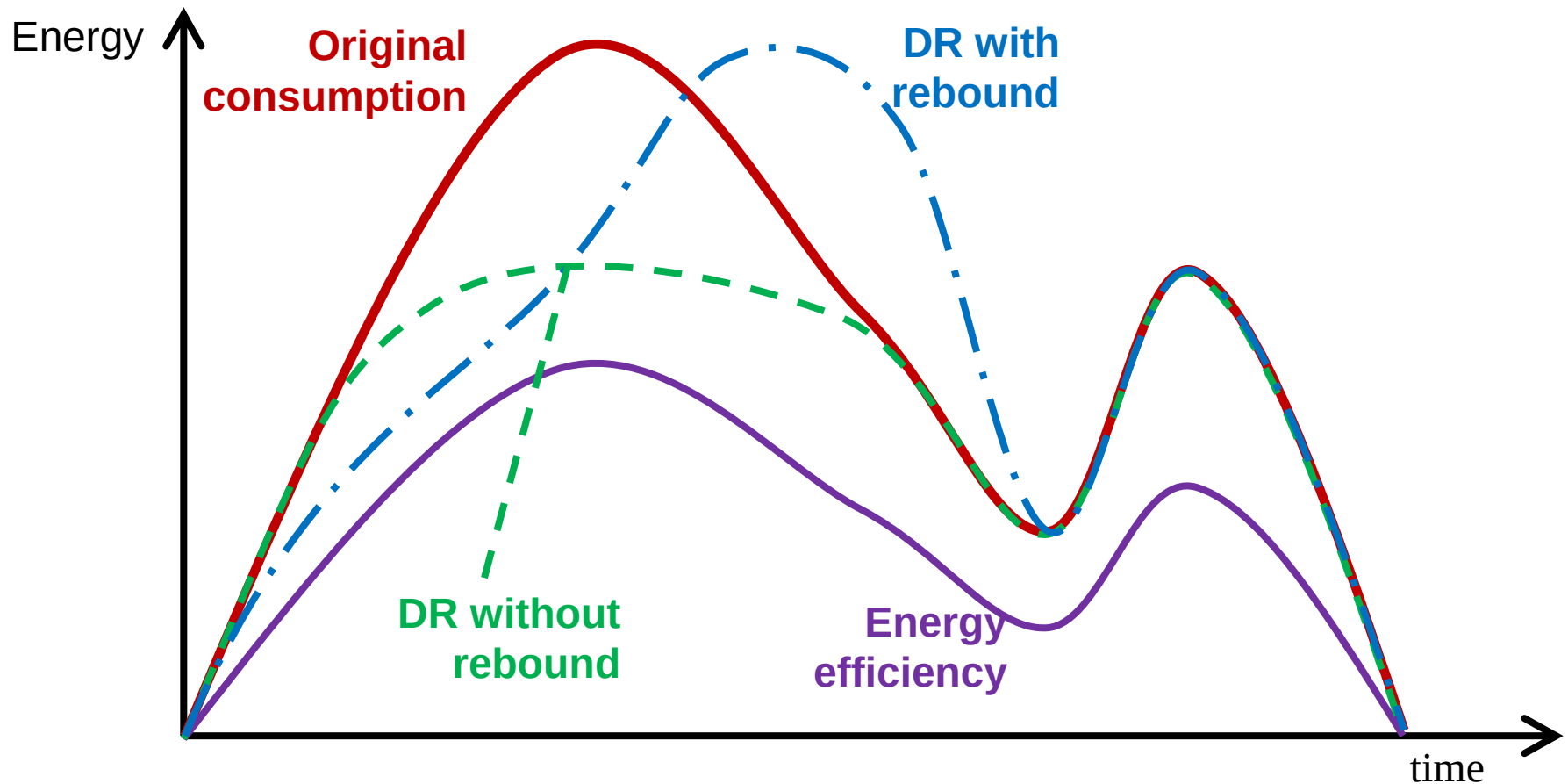
- ▶ What is Demand Response (DR)?
 - ▶ DR is the automatic voluntary or mandatory curtailment of electricity usage by utility and consumers at critical times when electricity demand threatens to outstrip capacity or in response to high electricity price
- ▶ Definition from US Department of Energy (DOE)
 - ▶ *“Changes in electric usage by end-use consumers from their normal consumption patterns in response to changes in the price of electricity over time, or to incentive payments designed to induce lower electricity use at times of high wholesale market prices or when system reliability is jeopardized.”*
- ▶ Demand Side Management (DSM) is a broader strategic approach encompassing long-term planning and implementation. DR can be considered a subset of DSM, representing the dynamic/responsive component within the broader DSM framework.

Drivers for Demand Response

▶ Drivers

- ▶ Skyrocketing prices in high demand periods in deregulated markets
- ▶ Load growth (particularly air-conditioners) without investing in new generation and/or delivery capacity
- ▶ Reducing risk of blackouts and brownouts during extreme events
- ▶ Avoiding or deferring costly infrastructure investments
- ▶ Government policies and measures to improve energy and market efficiency
- ▶ Policies encouraging more active consumer participation

Impact of improved energy efficiency vs demand response

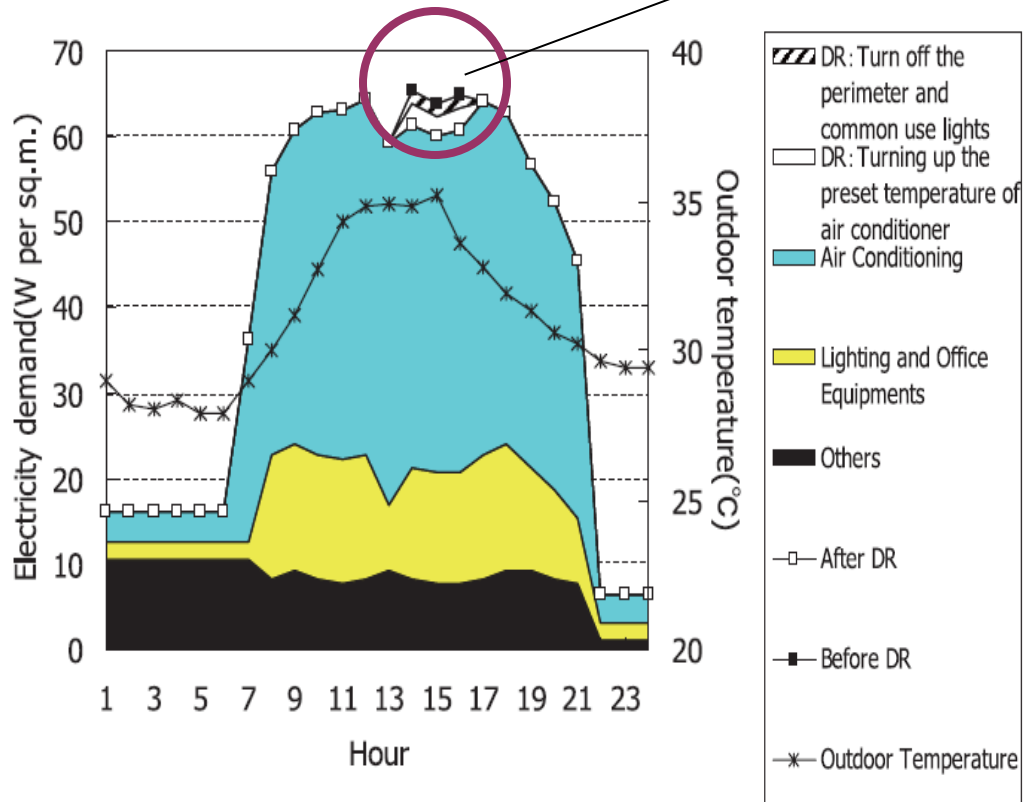


Energy Efficiency

- ▶ Improving energy efficiency of buildings or industrial sites starts with information and insight into the processes involved.
- ▶ For example, adjust the temperature of AC of a building to a higher number; or switch to more efficient appliances.



Demand Response



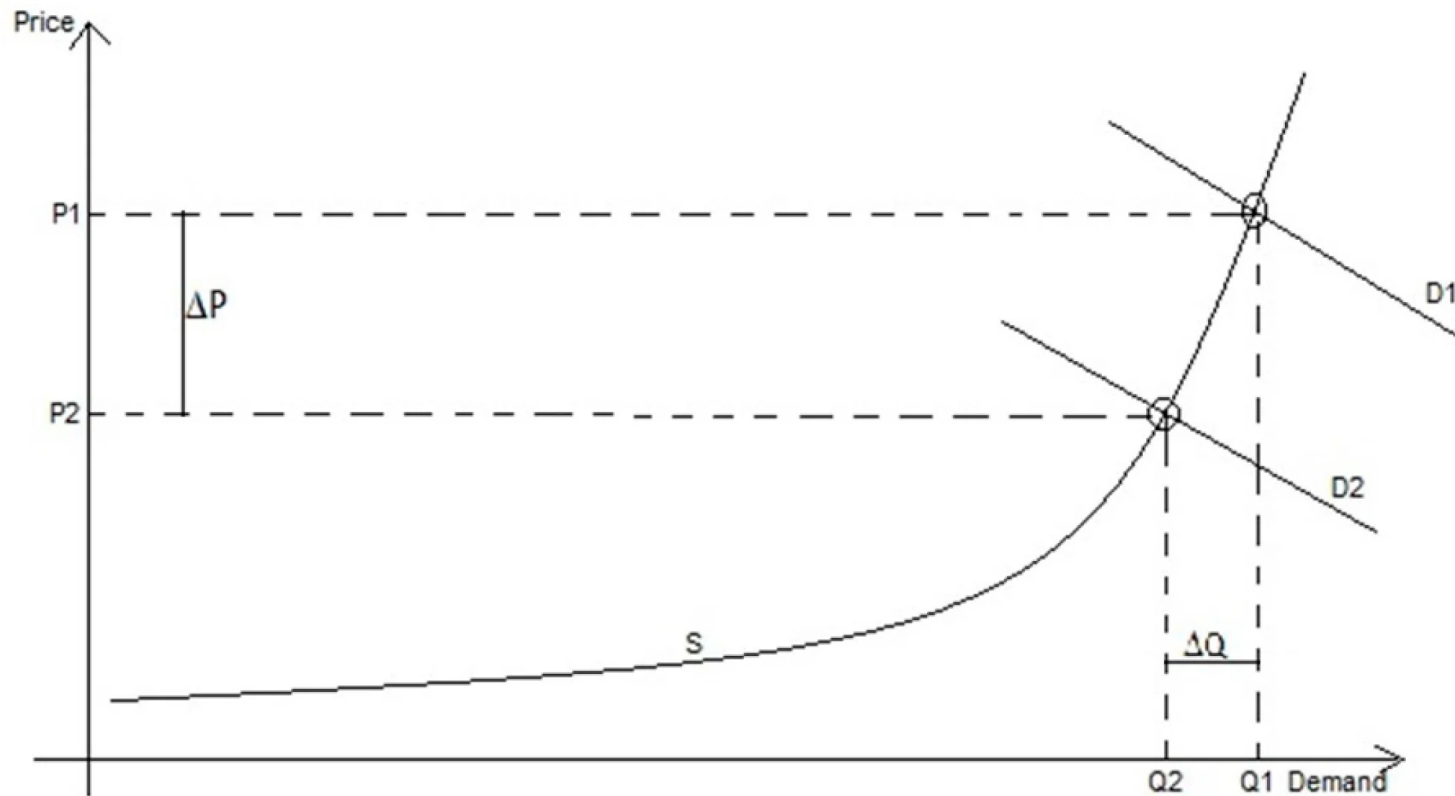
Reduce consumption by means of:

- 1) Bi-lateral agreement
- 2) Third party contracts
- 3) Special tariff structure
- 4) Smart appliances

Benefits:

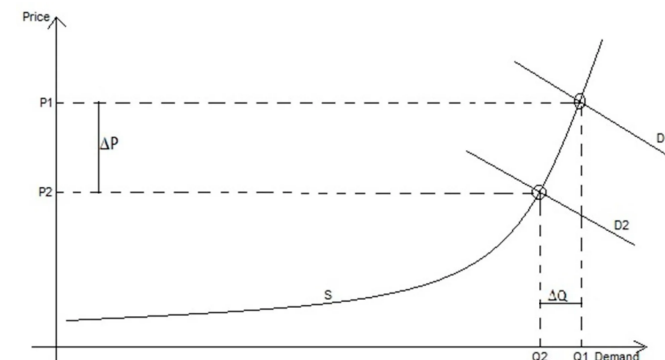
- ❖ cost savings
- ❖ investment deferral
- ❖ operation optimization
- ❖ service reliability

Demand Response & Electricity Price



Demand Response & Electricity Price

- ▶ When electricity is produced, generators are activated in order of cost - cheapest ones first, more expensive ones only as needed to meet demand. Without demand response:
 - ❖ Electricity demand is rigid (inelastic) - D1
 - ❖ The price is determined by the most expensive generator needed - P1
 - ❖ This creates high prices during peak demand
- ▶ With demand response:
 - ❖ Consumers can adjust their usage based on price signals
 - ❖ This makes demand more flexible (elastic) – D2
 - ❖ When prices rise, some consumption shifts to other times
 - ❖ This reduces the need for expensive peak generators – P2
 - ❖ The result is a lower overall electricity price



Types of Demand Response

Incentive based DR

- ▶ Direct load control (DLC)
 - ▶ Utility or grid operator gets free access to customer process
- ▶ Interruptible rates
 - ▶ Customers get special contract with limited sheds
- ▶ Emergency demand response programs
 - ▶ Voluntary response to emergency signals
- ▶ Capacity market programs
 - ▶ Customers guarantee to pitch in when the grid is in need

Time Rate based DR

- ▶ Time-of-use (TOU) rates
 - ▶ A static price schedule is applied
- ▶ Critical peak pricing (CPP)
 - ▶ A less predetermined variant of TOU
- ▶ Real-time pricing (RTP)
 - ▶ Wholesale market prices are forwarded to end customers

Direct Load Control (DLC)

▶ Function

- ▶ DLC programs refer to those in which a utility or system operator remotely shuts down or cycles a customer's electrical equipment on short notice to address system or local reliability contingencies in exchange for an incentive payment or bill credit.

▶ Earliest time for commercialization : 1980s

▶ Basic principle

- ▶ Cycles the operation of appliances such as air conditioners or water heaters

Interruptible / Curtailable (I/C) Rates

▶ Function

- ▶ Customers, by agreeing to reduce load during system contingencies on I/C rates, receive a rate discount or bill credit
- ▶ Customers can be penalized if they do not curtail
- ▶ Generally filed with regulatory commissions and offered to a utility's largest customers

▶ Sizes

- ▶ Range from 200kW (base interruptible program in California) to 3MW (American Electric Power's Ohio service territory)

▶ Customers type

- ▶ Not good candidates for customers with 24/7 operations or continuous processes such as schools, hospitals or hotels

Emergency Demand Response

▶ Function

- ▶ Provide incentive payments to customers for reducing their loads during reliability-triggered events, but curtailment is voluntary (i.e. customers can choose to forgo the payment and not curtail when notified without penalty)
- ▶ Mostly associated with Independent System Operator / Regional Transmission Organizations (ISO/RTO)

▶ Disadvantage

- ▶ Cannot accurately forecast how much load curtailment will occur when an event happens

Capacity Market Programmes

▶ Function

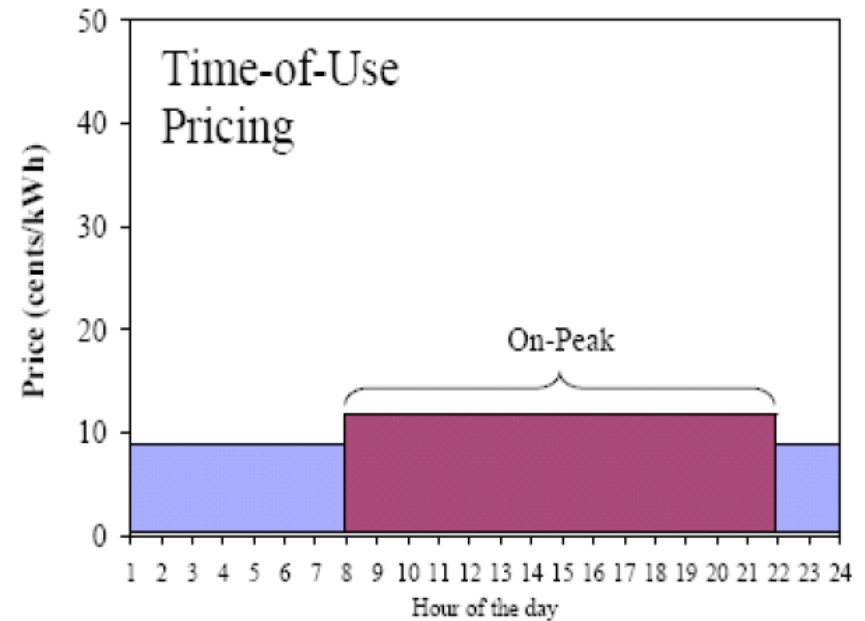
- ▶ Customers commit to providing pre-specified load reductions when an event happens, and are subjected to penalties if they do not curtail when directed
- ▶ Offered by wholesale market providers (e.g. ISOs/RTOs) and are preferred by curtailment service providers as well as customers because of the guaranteed payments

Ancillary Services

- ▶ Ancillary Services refer to specialized grid support functions that wholesale market providers can offer beyond basic energy supply. These services help maintain grid stability and reliability:
 - ▶ Rapid response (seconds to minutes) to small imbalances between generation and load to maintain grid frequency at 60Hz (US) or 50Hz (many other countries)
 - ▶ Allow customers to bid load curtailments in ISO/RTO markets as operating reserves
 - ▶ Maintaining voltage levels within acceptable ranges through reactive power management

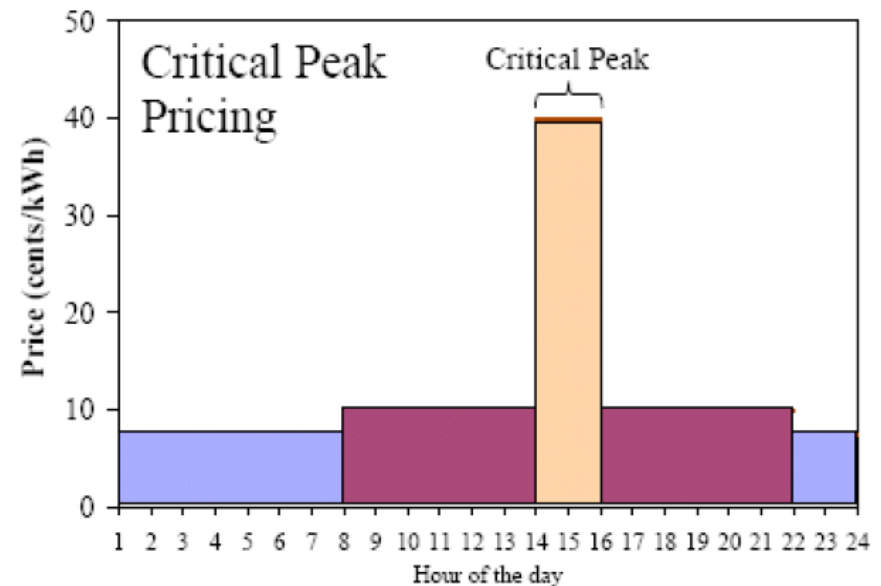
Time-of-use (TOU) rates

- ▶ TOU rates establish two or more daily periods that reflect hours when the system is higher (peak) or lower (off-peak), and charge a higher rate during peak hours
- ▶ Off-peak hours are usually some part of the evening and nights, as well as weekends



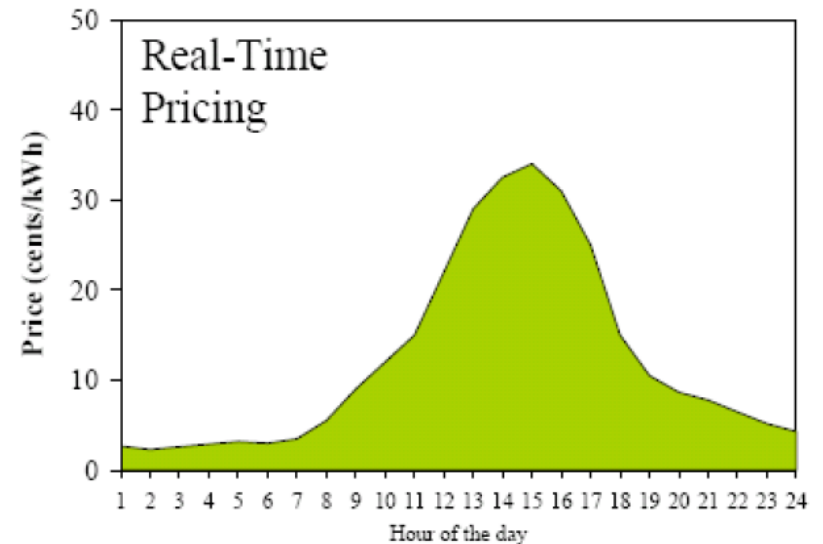
Critical peak pricing (CPP)

- ▶ CPP is an overlay on either TOU or flat pricing and uses real-time prices at times of extreme system peak
- ▶ May be triggered by system contingencies or high prices faced by the utility in procuring power in the wholesale market



Real-time pricing (RTP)

- ▶ RTP rates vary during the day, directly reflecting wholesale electricity prices, as opposed to rate designs such as TOU or CPP that are largely based on preset prices
- ▶ The first RTP programs were introduced in California in the mid-1980s
- ▶ Delaware, New Jersey, Pennsylvania, Maryland, Ohio, New York and Illinois have initiatives aimed at implementing default RTP for the largest customers



Popular type of DRs

More popular



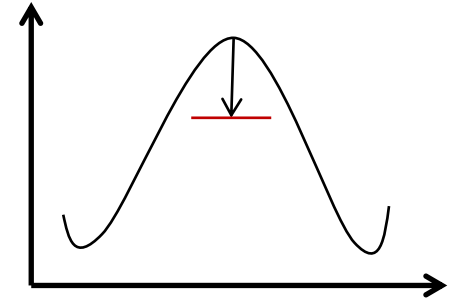
Direct Load Control
Interruptible / Curtailable
Time of Use
Real-time Pricing
Emergency Demand Response
Critical Peak Pricing
Capacity Market Program
Demand Bidding / Buyback
Ancillary Services

Less popular

Load Shape Objectives of Load Management

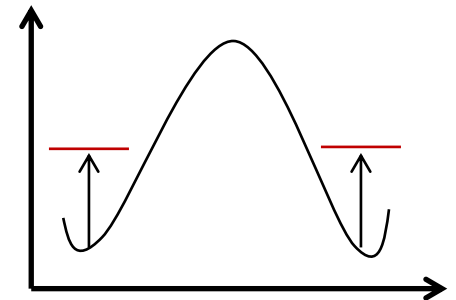
▶ Peak Clipping

- ▶ Generally use direct load control (DLC) to reduce the peak load
- ▶ To reduce operating cost and dependence on critical fuels by economic dispatch



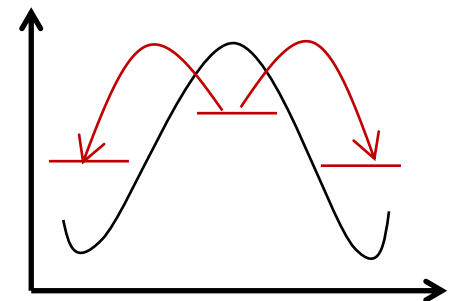
▶ Valley Filling

- ▶ Desirable where the long-run incremental cost is less than the average price of electricity
- ▶ Can be accomplished by adding new thermal energy storage (water heating)



▶ Load shifting

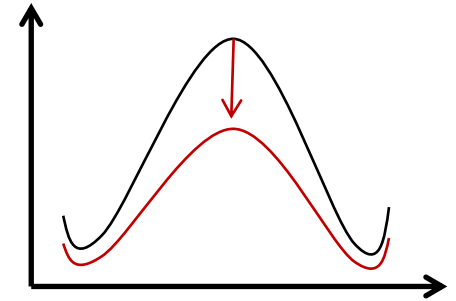
- ▶ Involves shifting load from on-peak to off-peak periods
- ▶ Popular applications include use of storage water heating, storage space heating and coolness storage and etc..



Load Shape Objectives of Energy Efficiency and DR

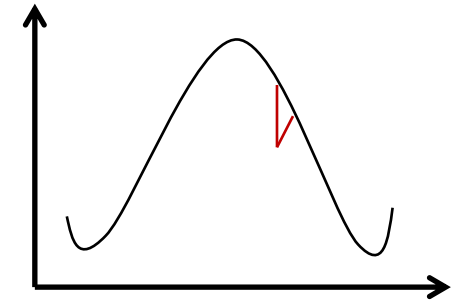
- ▶ **Energy Efficiency**

- ▶ Results from programmes directed at end-use consumption



- ▶ **Reliability Response (Contingency Response)**

- ▶ Requires the fastest, shortest duration response
- ▶ Response is only required during power system contingencies

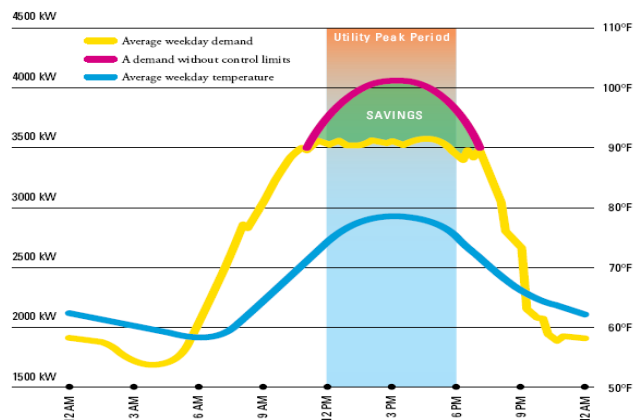


Demand Response example

- ▶ Location: Yahoo headquarters, California
- ▶ Seven-building campus with offices, café, fitness center and retail shops
- ▶ Installations: direct digital control system and high efficiency fluorescent lamps
- ▶ 15% energy savings from demand response

Savings resulting from daily peak period demand control

Source: PG&E Integrated Energy Audit, October 2004.



Building systems control room



Key Benefits of DR

▶ Participant financial benefits

- ▶ Bill savings and incentive payments earned by customers for changing their consumption patterns

▶ Market-wide financial benefits

- ▶ Drive down production costs and the wholesale prices by reducing the costly peaking units
- ▶ Reduce the need to build more capacity and all these savings can be passed on to consumers

▶ Reliability benefits

- ▶ Lower the likelihood and consequences of loss-of-load due to adverse conditions

▶ Market performance benefits

- ▶ Mitigate the suppliers' ability to exercise market power by lowering the demand

Pros and Cons of DR

Advantages



Avoid operating peaking units and/or defer the need to build them



Reduce service interruption risks and vulnerability



Reduce system incremental costs and overall price



Deter dominant market power



Energy savings

Disadvantages

- ▶ Requires smart meters and associated communication infrastructure to function
- ▶ Complicated DR management systems
- ▶ Benefits difficult to quantify and allocate

Enablers and Barriers of DR

Enablers



Dynamic electricity rates structure



Intelligent end use technologies (e.g. smart appliances & thermostats, micro CHP, energy storage devices, etc.)

Barriers

- ▶ Lack of regulatory initiative and coordination
- ▶ Slow deployment of smart meters and AMI
- ▶ Complications and high cost of integrating DR with local EMS
- ▶ Customer's fear of interruptions and annoyances
- ▶ Market barriers to load aggregators
- ▶ Lack of load entities suitable for participation in DR programs
- ▶ Inaccessibility of real-time data by market participants
- ▶ Slow pick-up and learning rate in customers



Demand Response in US ISOs



FERC Order 719 (2007)

- ▶ Federal Energy Regulatory Commission (FERC) is an Independent Regulatory Agency of the Federal Government
- ▶ Key elements of FERC Order 719
 - ▶ DR as a source of energy and ancillary service, including retail demand response
 - ▶ Scarcity pricing during reserve shortages
 - ▶ Support for long-term power contracting
 - ▶ Enhanced market-monitoring policies
 - ▶ Increased responsiveness of ISOs to their customers

FERC Order 745 (2010)

► Purpose of Order 745

- To require that DR resources participating in RTO or ISO markets are paid at the Locational Marginal Price (LMP) when
 1. Such resources contribute to the supply-demand balance as a substitute for generation and;
 2. demand response resources pass a net benefits test accepted by FERC

What are ISO and RTO?

- ▶ Independent System operator (ISO) is an organization to coordinate, control and monitor the operation of the power grid (usually within a single U.S. state)
- ▶ Regional Transmission operator (RTO) also coordinates, controls and monitors the power grid over a wider area that crosses state borders
- ▶ ISOs and RTOs manage the complex interconnections
- ▶ ISOs and RTOs are also required to maintain orderly wholesale marketplaces that enable participants buy and sell electricity on a day-ahead or a real-time spot market

ISO/RTO and DR

- ▶ DR programs have traditionally been run by utilities
- ▶ ISOs and RTOs today run their own demand response programs, and are increasingly factoring DR resources in their capacity and transmission planning

Summary of U.S. RTO/ISO DR Programs and Rules

- ▶ Based on energy and ancillary service markets facilitated by the ISOs/RTOs, DR programs can be broadly classified as
 - ▶ Market-based (via economic signals)
 - ▶ Reliability-based (triggered under emergency conditions)
- ▶ The market products DR can offer
 - ▶ Capacity (installed capacity, or UC availability in some markets)
 - ▶ Energy (day-ahead, real-time balancing)
 - ▶ Reserve ancillary services
 - ▶ Regulating reserve
 - ▶ Spinning reserve/responsive reserve
 - ▶ Supplemental reserve/non-spinning reserve

US ISO – PJM

- ▶ PJM Interconnection is a RTO that coordinates the movement of wholesale electricity in all or parts of 13 states and the District of Columbia
- ▶ DR participation in PJM's energy markets include
 - ▶ Economic load response, which may be provided by agent PJM members, or Curtailment Service Providers (CSPs). It is an economic program that uses the LMP as a trigger. Consumption under this program is curtailed when $LMP \geq \$75/\text{MWh}$
 - ▶ A three-year forward capacity auction, called the Reliability Pricing Model (RPM), in June 2007
 - ▶ RPM allowed DR to compete on equal footing with traditional generators as capacity resource

US ISO – ISO-NE

- ▶ ISO New England (ISO-NE) is an independent, not-for-profit corporation created in 1997 to oversee New England's bulk electric power system
- Real-time Demand Response (RDR) and Real-time emergency generation (RTEG) Program

Real-Time Price Response Program in ISO-NE

- ▶ Customers that participate in the Real-Time Price Response Program (RTPRP) act on a strictly voluntary basis. These customers:
 1. Utilize price signals to decide whether to voluntarily reduce load
 2. Will be notified when the forecast hourly Zonal Price is greater than or equal to \$100/MWh on a Monday-Friday, non-Demand Response Holidays between 7:00 AM – 6:00 PM
 3. Once notified, the window of availability for Real-Time Price Response can be as early as 7 AM and remain open until 6 PM (i.e., between the hour ending 0800 through the hour ending 1800)
- ▶ Enrolling Participant will receive the higher of the applicable Real-Time Zonal Price for interrupted consumption (measured against the base line) or a guaranteed minimum payment of \$100/MWh when the eligibility period is open
- ▶ RTPRP Demand Resources that fail to respond when the ISO opens the window to curtail load are not subject to any penalties

Calculation Examples of Price Response

- ▶ Situation: 4.5 MW reduction commitment by customer and customer interrupts when called upon
- ▶ Payments: The following payments would be made to the Enrolling Participant:
 - ▶ Minimum Energy Payment for Price Response equals \$100 per MW interrupted
 - ▶ LMP for Hour 8 of interruption = \$92.00/MWh
 - ▶ LMP for Hour 9 of interruption = \$100.35/MWh
 - ▶ Participant delivers 2.5 MW (7.000 - 4.500) in hour 8 and 4.01 MW (7.000 – 2.990) in Hour 9.
 - ▶ LMP payments to participant are:
 - ▶ For Hour 8: 2.5 MWh * (\$100.00) = \$250.00
 - ▶ For Hour 9: 4.01 MWh * (\$100.35) = \$402.40
 - ▶ Total payment for interruption = \$652.40

Hour	Initial Profile (MW)	Adjusted Profile (MW)	Metered load (MW)	Locational Marginal Price Per Hour (\$/MWh)	Minimum Payment (\$/hr)
Hour 8	6.500	7.000	4.500	92.00	100.00
Hour 9	6.500	7.000	2.990	100.35	100.00

US ISO – ERCOT (I)

- ▶ The Electric Reliability Council of Texas (ERCOT) operates the electric grid and manages the deregulated market for 75 percent of the state
 - ▶ Serves 85% of Texas' load
 - ▶ Possess the “peakiest” demand profile of all the ISOs in US
- ▶ Tariff-based program - “4 coincident peak” (4CP) program
 - ▶ The demand of large industrial and commercial customers is measured in 15-minute intervals during the summer months' four peak days. Transmission charges are then calculated based upon a customer's peak demand during 4CP periods
 - ▶ Sophisticated energy consumers will attempt to predict 4CP periods and drop their load accordingly to save roughly \$22,000/MW-year

US ISO – ERCOT (II)

- ▶ “Load acting as a Resource” (LaaR): ERCOT’s largest DR program with 2.38 GW
 - ▶ Curtailments are manually controlled in response to grid operator dispatch with 1.15GW cap
 - ▶ LaaR payments decreased from \$270m in 2008 to \$90m in 2010 due to a drop in energy prices resulting from cheap natural gas and the recession’s effect on electricity demand
- ▶ “Emergency interruptible load service” (EILS): ERCOT’s second DR program with 475MW
 - ▶ Loads are contracted to curtail during emergencies for up to eight hours during four-month time periods

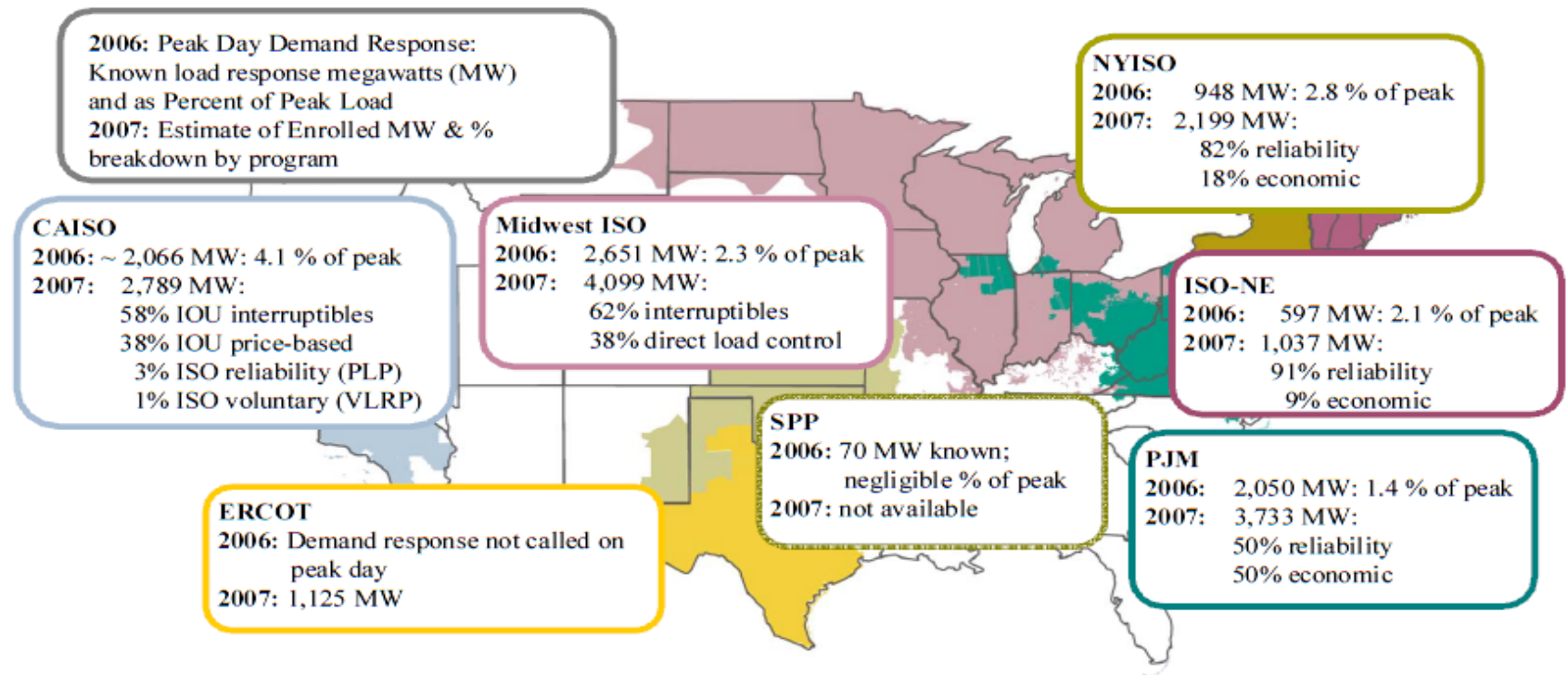
US ISO – NYISO (I)

- ▶ New York Independent System Operator (NYISO)
 - ▶ Two reliability-based programs
 - ▶ Emergency Demand Response Program (EDRP)
 - ▶ Installed Capacity - Special Case Resources (ICAP-SCR)
 - ▶ Two economic-based programs
 - ▶ Day-ahead Demand Response Program (DADRP)
 - ▶ Demand-side Ancillary Service Program (DSASP)
- ▶ EDRP is only called upon during grid emergencies
 - ▶ Participation in this program is voluntary and there are no penalties for not responding

US ISO – NYISO (II)

- ▶ ICAP-SCR is the largest program by revenue in NYISO and makes up 80% of the total DR portfolio
 - ▶ Capacity-based, composed of 90% load and 10% behind-the-meter generation
 - ▶ Has been called upon to curtail 22 times for periods ranging 4-9 hours in ten year of operation
- ▶ Of the two economic programs (DADRP and DSASP), only DADRP holds participants, totaling 331 MW

Summer 2006 DR Contributions and Summer 2007 Program Enrollments



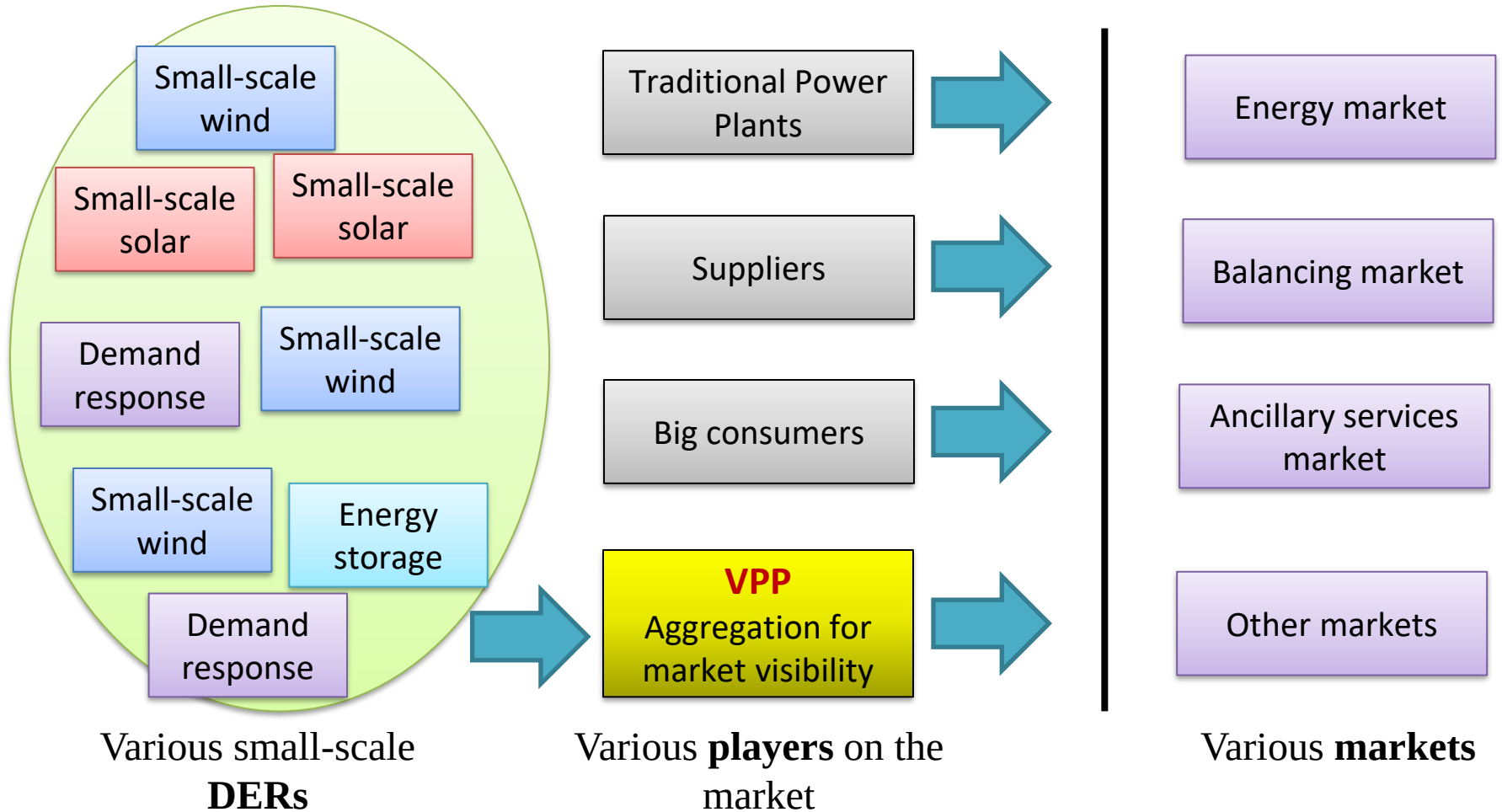
United States ISOs and RTOs	Regions Covered
New York Independent System Operator (NYISO)	NY
ISO New England (ISO-NE)	CT, ME, MA, NH, RI, VT
PJM Interconnection LLC (PJM)	DE, IL, IN, KY, MD, MI, NJ, NC, OH, PA, TN, VI, WV, DC
Southwest Power Pool (SPP)	AR, KS, LA, MS, MO, NM, OK, TX
Electric Reliability Council of Texas (ERCOT)	TX
California Independent System Operator (CAISO)	CA
Midwest Independent Transmission System Operator	MT, ND, SD, NE, MN, IA, IL, MO, WI, MI, IN, OH, AR, PA, WV



Virtual Power Plant



Overview of Virtual Power Plant (VPP)



Virtual Power Plant : technology for market visibility of many small DERs

DER considered for integration in VPP

- ▶ DER includes not only positive power generation such as PV, wind, and micro-turbines, but also negative power generation through demand response programmes, controllable loads, and direct load control. Possible DERs for VPP:
 - i. CHP (Combined Heat and Power)
 - ii. Small power plants (gas turbines, diesel generators, micro-turbine etc.)
 - iii. Small Hydro-plants
 - iv. Small-scale wind
 - v. Small-scale solar (e.g. roof top PVs)
 - vi. Demand response (with controllable/dispatchable loads)
 - vii. Energy storage systems

Why VPPs are needed?

- ▶ VPP brings small DER on the market
 - ▶ Stand alone DER electrical power and potential services are usually below efficient market thresholds
- ▶ Some DER have unpredictable or low-predictable behavior due to external factors (wind, solar)
 - ▶ they need to be correlated with a balancing responsible party in order to cover unbalance
- ▶ Existing balancing responsible party are not active, so they are more administrating rather than optimizing the unbalance

Where VPP can help ?

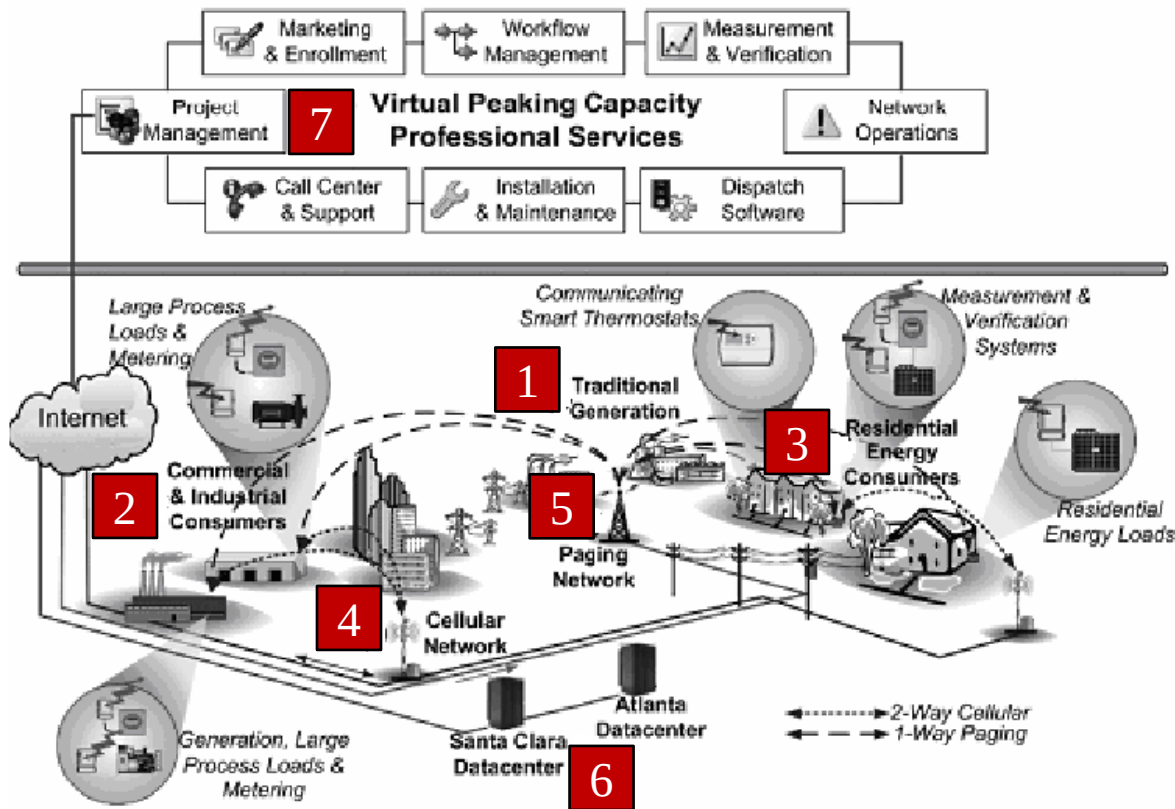
DERs shall be integrated in a smarter way

Enable opening to the energy markets for small producers (energy and ancillary/network services)

Improve the security of supply in the situation of high penetration of uncontrollable generation

Alternative to network reinforcement

Comverge Virtual Peaking Capacity (VPC) Power Schematic



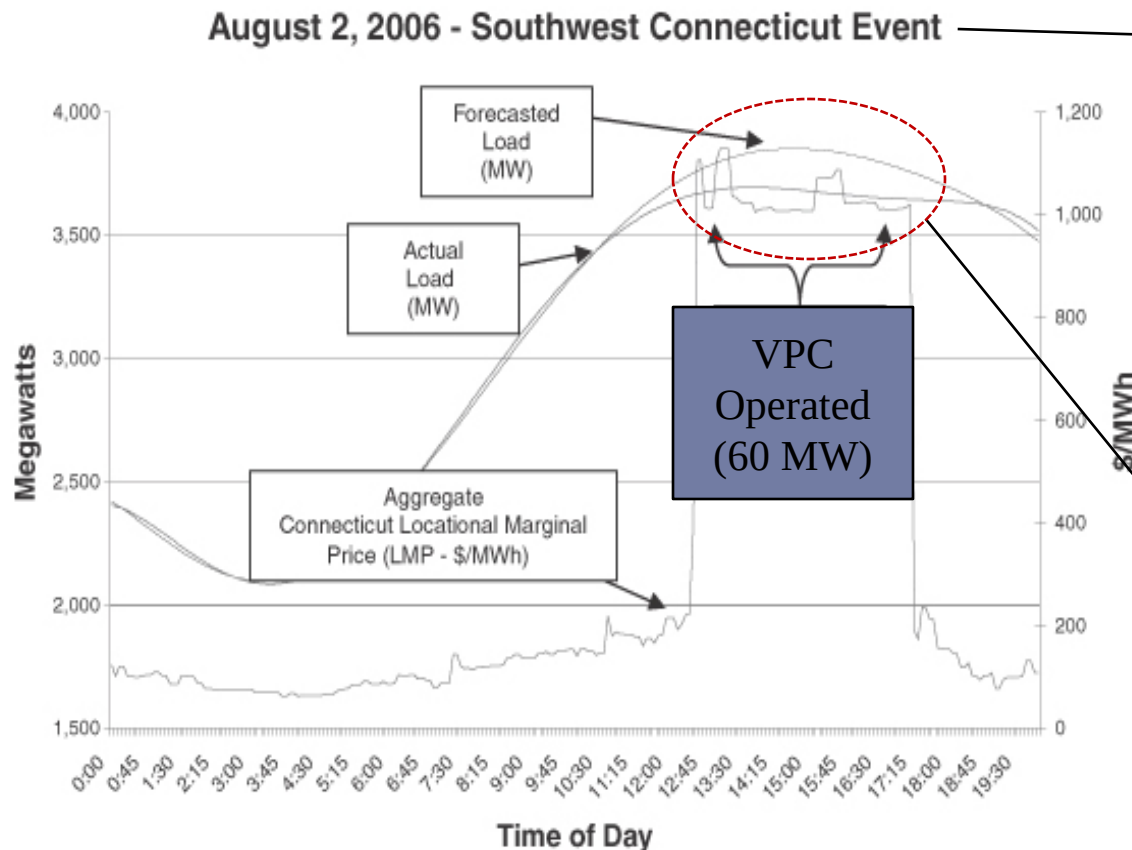
Traditional generation sources (1) provides electricity to commercial and industrial (2) and residential (3) energy consumers

Control of the equipment is accomplished by sending signals to the control equipment using cellular (4) and paging (5) networks

The communication networks are operated from software systems and redundant data center facilities (6)

Using the internet and the applications operating at the data centers, Comverge's internal staff and strategic vendors conduct many professional services (7) to fulfill contractual requirements, ensure reliability and meet the operational requirements required of its customers

Effect of a typical DR Programme run by Comverge



It illustrates the effect of the VPC program on reducing and shifting the overall load in a Southwestern Connecticut DR event compared to the forecasted load

A distinct reduction of the electricity demand from forecast during a period where the LMP was over \$1,000/MWh



Smart Meters and AMI



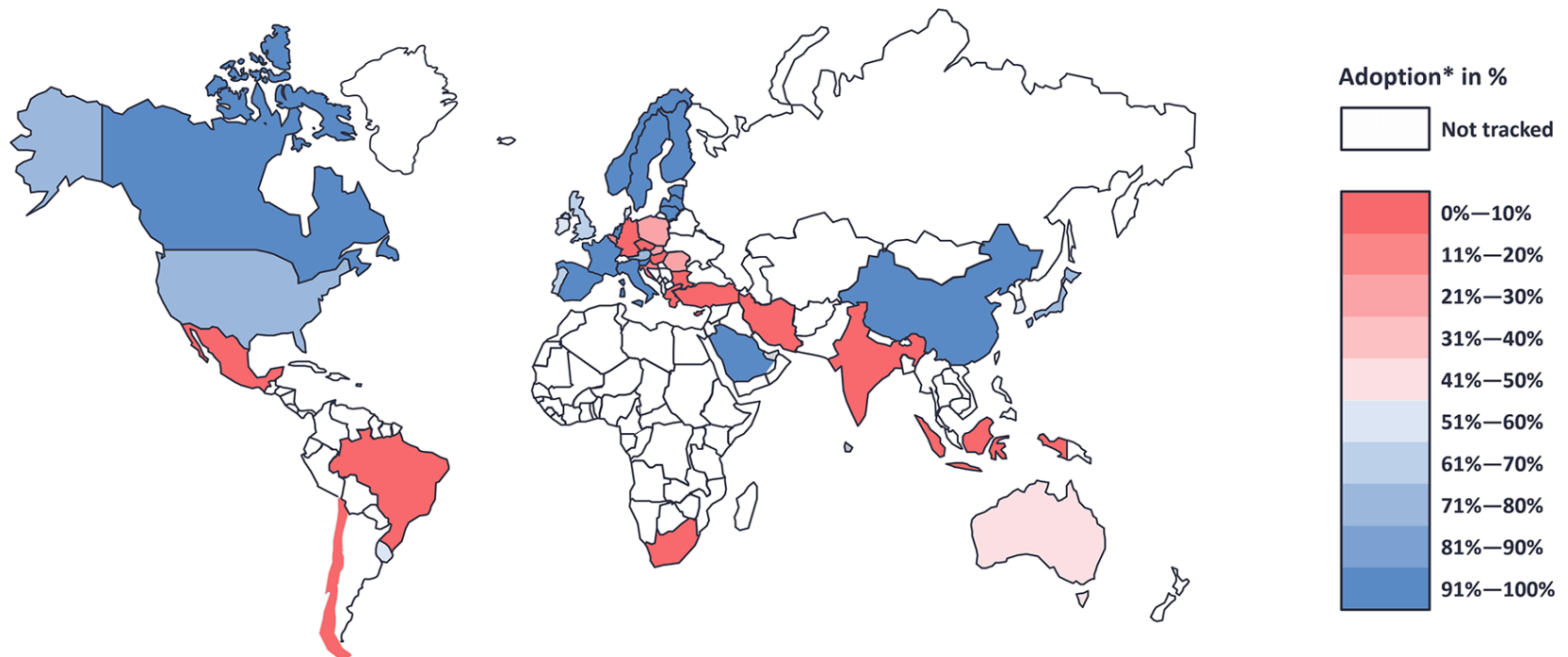
Smart Meters

- ▶ An advanced meter which not only record detailed consumption information but also support additional capabilities such as Automatic Meter Reading (AMR), bi-directional communication, TOU tariff support, in-home display, net-metering, pre-payment, remote connect/reconnect, smart appliance controls, power quality and outage monitoring



Global smart meter installation

Global smart electricity meter adoption 2024



* =Smart meter adoption (as of Q4 2023) = The share of smart (AMI) electricity meters installed as a share of the total number of metering points that includes residential, commercial, and industrial segments.
Source: IoT Analytics Research 2024 – Global Smart Meter Market Tracker 2020-2030. We welcome the republishing of images but ask for source citation with a link to the original post or company website.

How Smart Meter Works?

SM provides intelligent functions which include detailed recordings of energy usage with accurate time-stamps for:

- TOU-tariffs
- In-home display unit for customers to monitor their own usage
- Two-way communication for dynamic interaction capability
- Net-metering for integrating distributed resources (e.g. solar panels etc.)
- Prepayment meters for revenue protection
- Remote connect/disconnect capability for better service

SM require the support of advanced communication and IT systems to help deliver its full capabilities

Summary of Overall Cost Benefits by Case Study of Smart Meter

- ▶ Some cost benefit analyses conducted by different utilities, government and vendors are summarized
- ▶ Most of these studies included the benefits derived from automatic metering reading, TOU schemes and different levels of demand side management. It is apparent that their results differ although most of them show a positive Net Present Value (NPV) if SM is implemented.

Key Financial Data	Vic	PG&E	SCEd	Ofgem	SDGE	Bayard	OTTER
Benefit Cost Ratio	1.48	1.01	0.59	1.34	1.08	2.03	0.50
Benefits per customer A\$ NPB	\$474	\$529	\$218	\$295	\$385	\$696	\$309
End-point hardware Net Present Costs SA/meter	\$125	Not known	\$141	\$131	\$115	\$179	Not known
Total Net Present Costs SA/meter (including end-point hardware)	\$321	\$525	\$368	\$220	\$357	\$343	\$616
Net Present Value	\$153	\$4	-\$150	\$75	\$28	\$353	-\$307

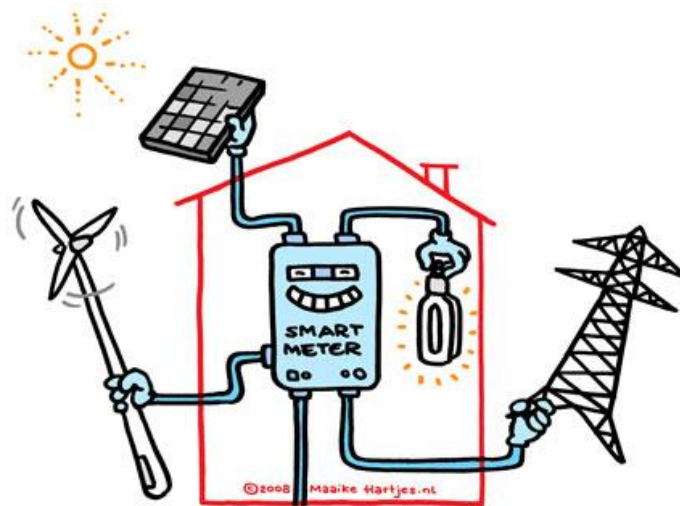
PG&E, SCEd and SDGE are California's major utilities

Ofgem is the regulator in UK and Vic is the Victoria government in Australia

Bayard is a Australian-based investment company who owns one of the largest SM supplier Landis & Gyr

Demand response and smart meters

- ▶ Demand response will increase with more “smart meters”
- ▶ Electricity users will become more aware of the cost of electricity with actual price signals
- ▶ Allow for distributed storage with EVs



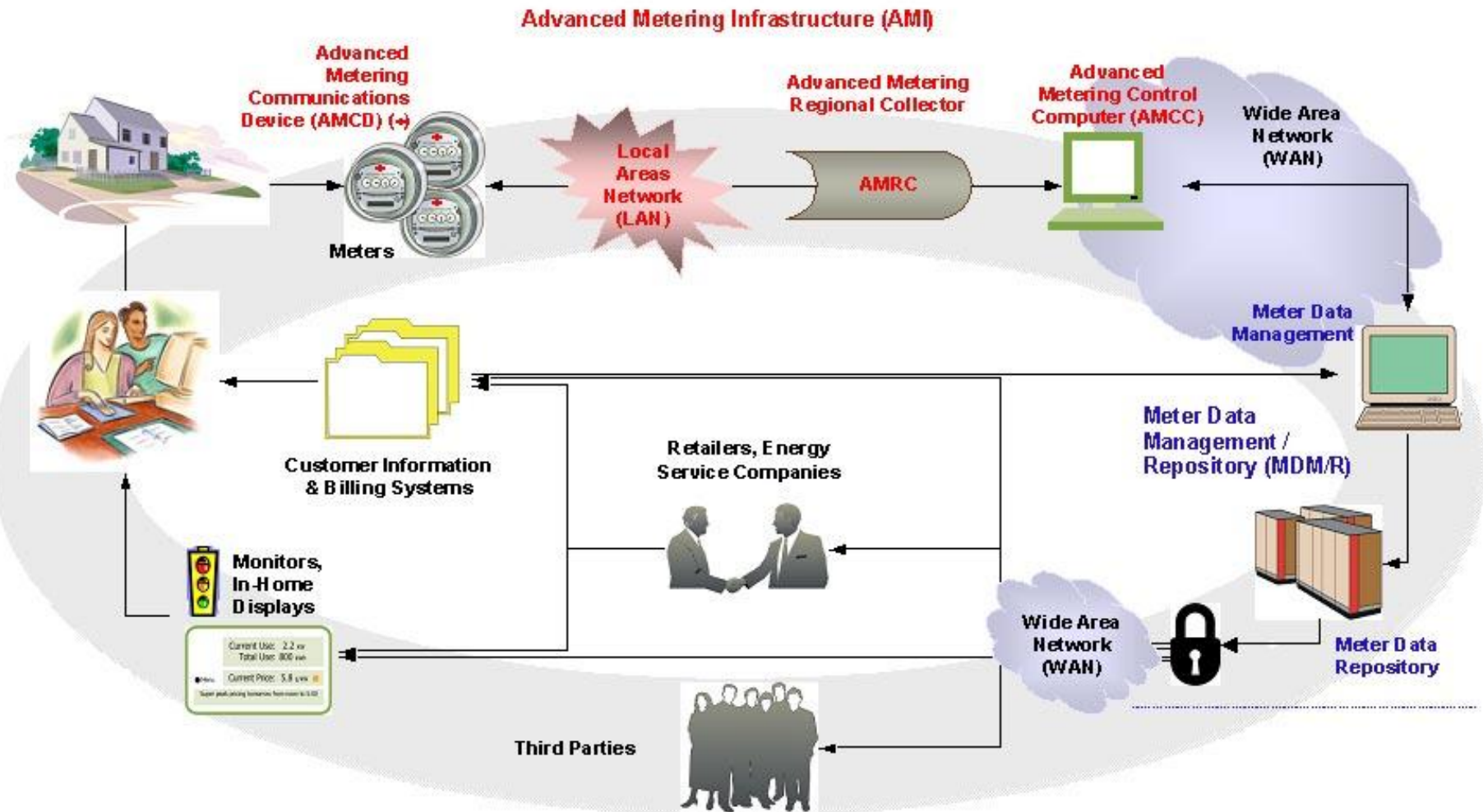
Advanced Metering Infrastructure (AMI)

- ▶ Advanced Metering Infrastructure (AMI) is similar to an Automatic Meter Reading (AMR) system but is equipped with bi-directional and high-speed communication capabilities
- ▶ Detailed usage data are collected on a regular basis (e.g. every 15 minutes or higher) by the smart meters and real-time price signals or interactive data can be sent/received between the smart meters and the service provider over a communication network

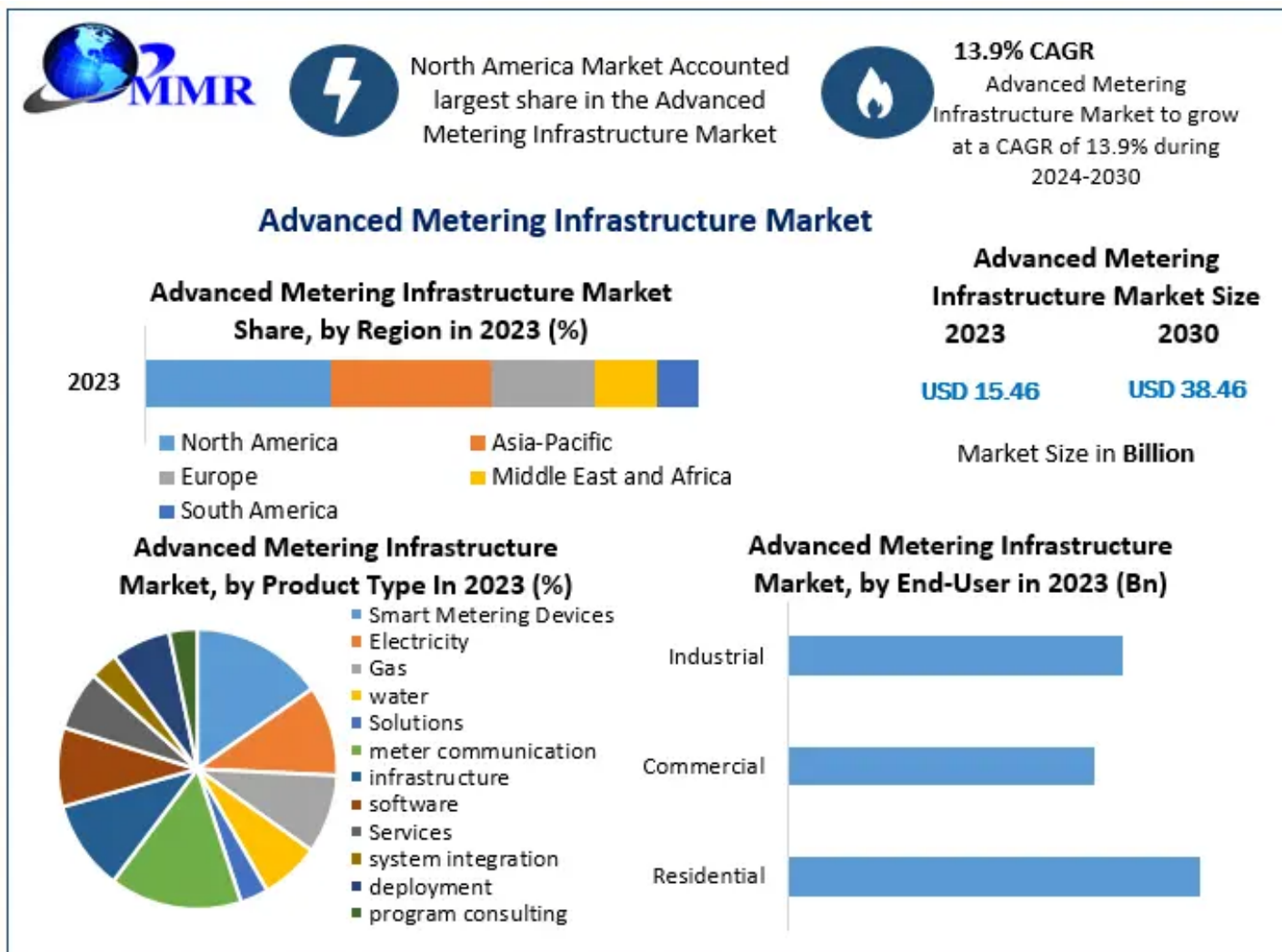
AMI Data Collection Network Options

- ▶ **Broadband over power line (BPL)**
 - ▶ BPL works by modulating high-frequency radio waves with the digital signals from the Internet
- ▶ **Power line communications (PLC)**
 - ▶ PLC systems send data through power lines by injecting information into either the current, voltage or a new signal
- ▶ **Fixed RF systems**
 - ▶ In basic fixed radio frequency (RF) systems, meters communicate over a private network using RF signals
- ▶ **System utilizing public networks**
 - ▶ These systems utilize existing public networks such as paging, satellite, Internet and/or telephony (cellular or landline) networks to provide for communication between meters and utilities

Smart Meters & AMI



AMI world-wide market





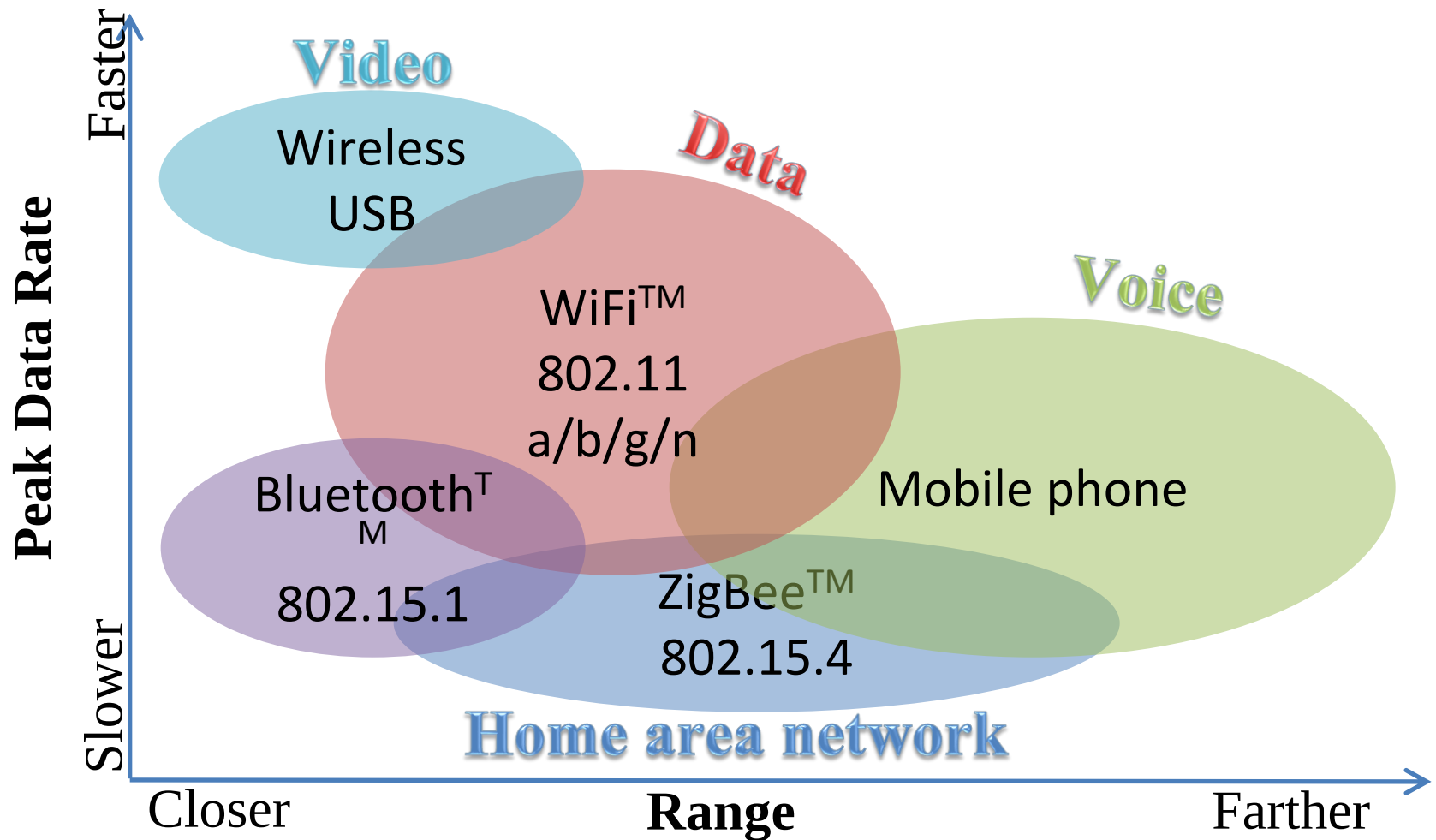
Home Area Network and Smart Home



Home Area Network (HAN)

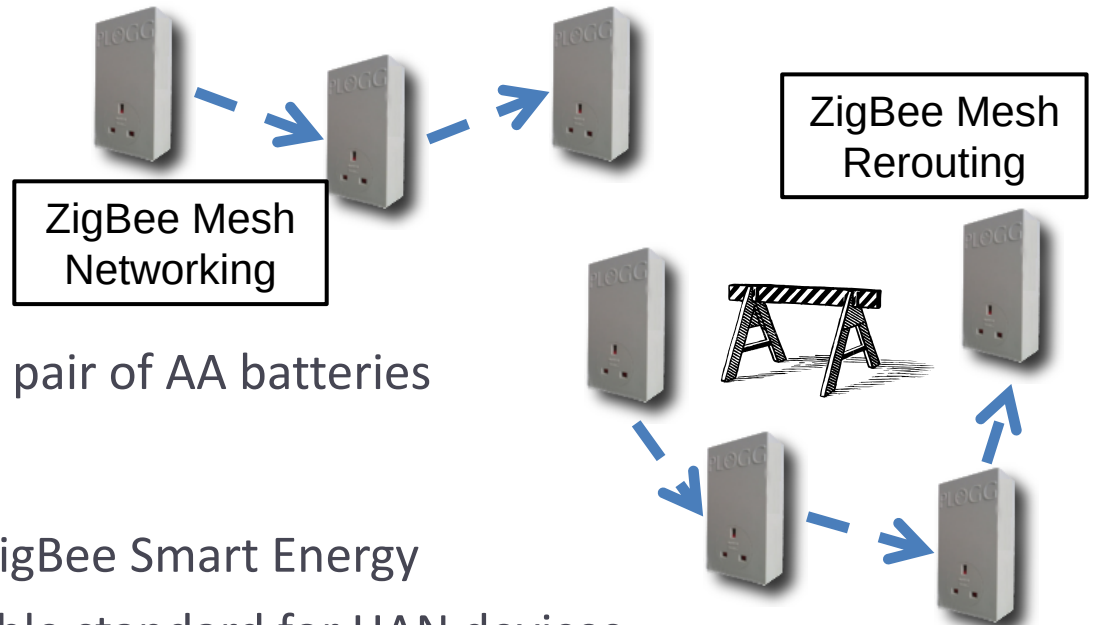
- ▶ Home Area Network (HAN) extends smart grid capabilities into the home using different networking protocols.
- ▶ HAN is the backbone of the communication between smart meter and home appliances
- ▶ A network portal or gateway that connects one or more outside information services to the HAN.

Wireless Technologies Comparison



Why ZigBee?

- ▶ Highly Reliable
- ▶ Cost effective
- ▶ Low-power
 - ▶ Can achieve 5 years on a pair of AA batteries
- ▶ Interoperability
 - ▶ NIST and DOE selected ZigBee Smart Energy as an initial interoperable standard for HAN devices



News Analysis

Home area networks: Zigbee beats WiFi in cost, says GE

Dec 9, 2010



08 December 2010

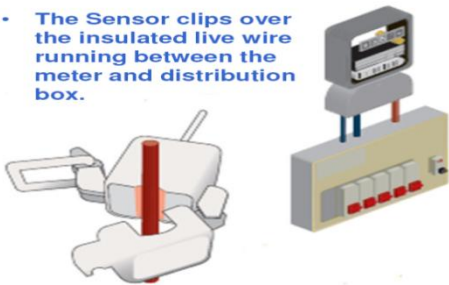
GE's Nucleus™ Energy Manager is First Energy Gateway to be ZigBee®-Certified

Energy Monitoring System

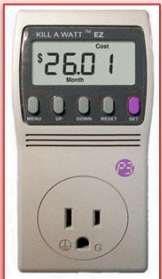
Meter-level

Connecting the Sensor

- The Sensor clips over the insulated live wire running between the meter and distribution box.



Plug-level

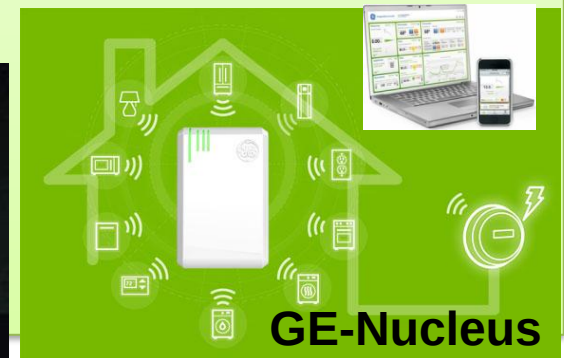
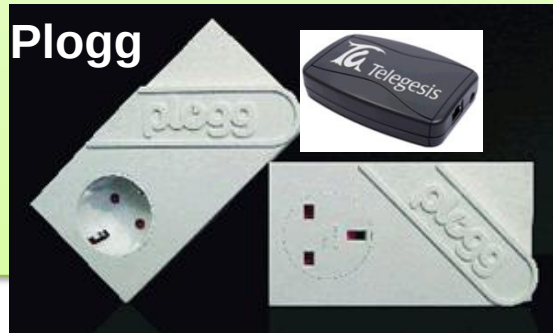


Meter + Plug + Energy monitoring system

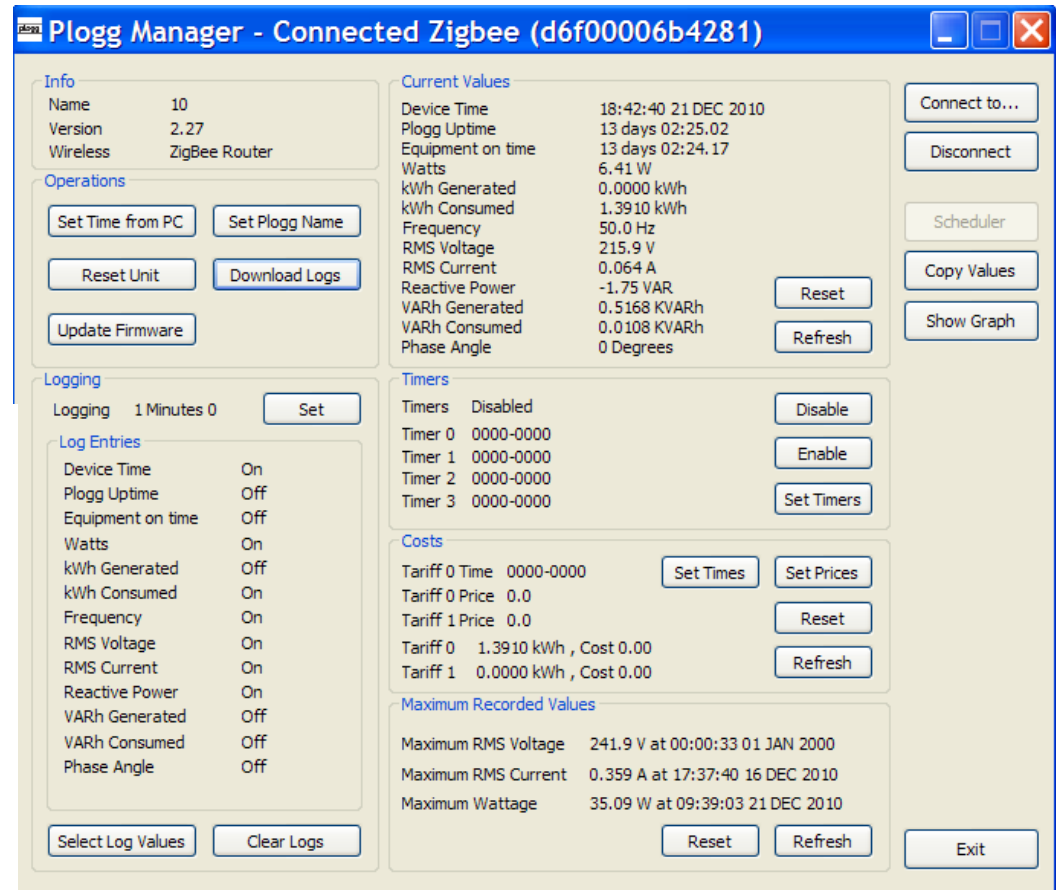
Alert me – Energy control



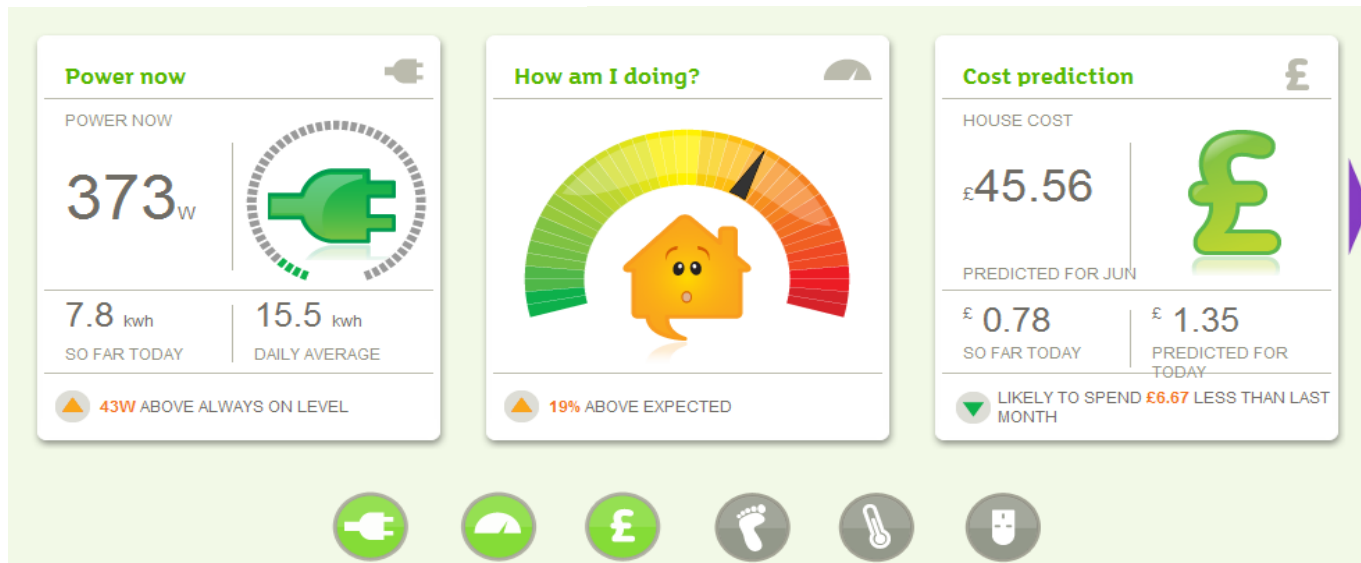
Plogg



Commercial Smart meter plug - Plogg



AlertMe - Home Energy Management



CISCO – Energy Management

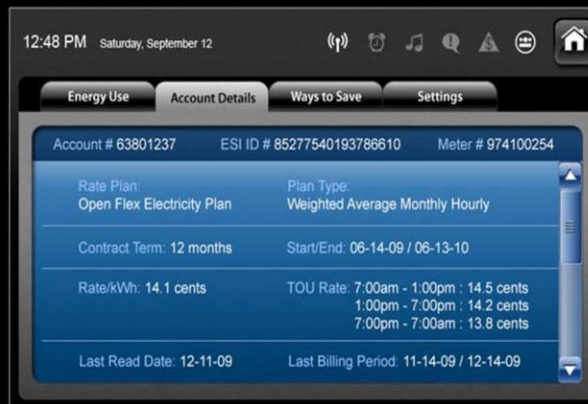
Energy Use



Thermostat

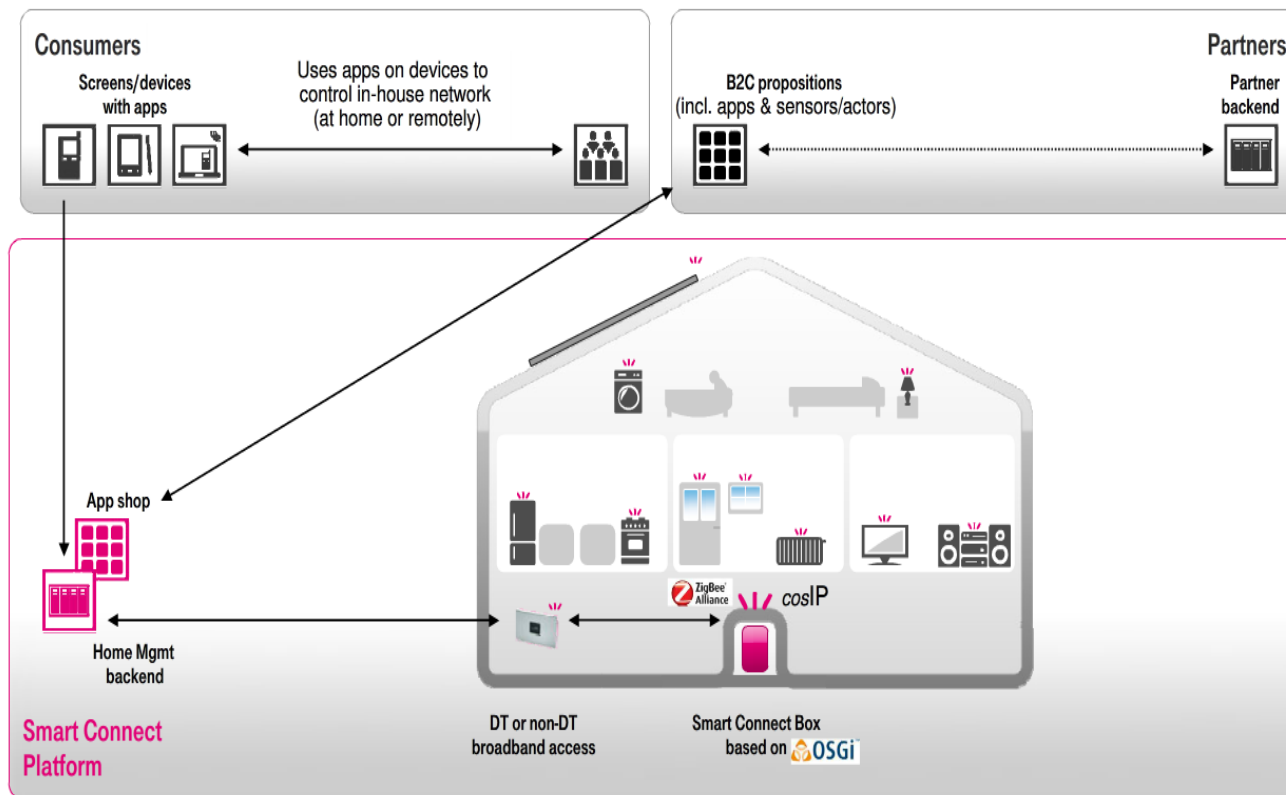


Ways To Save



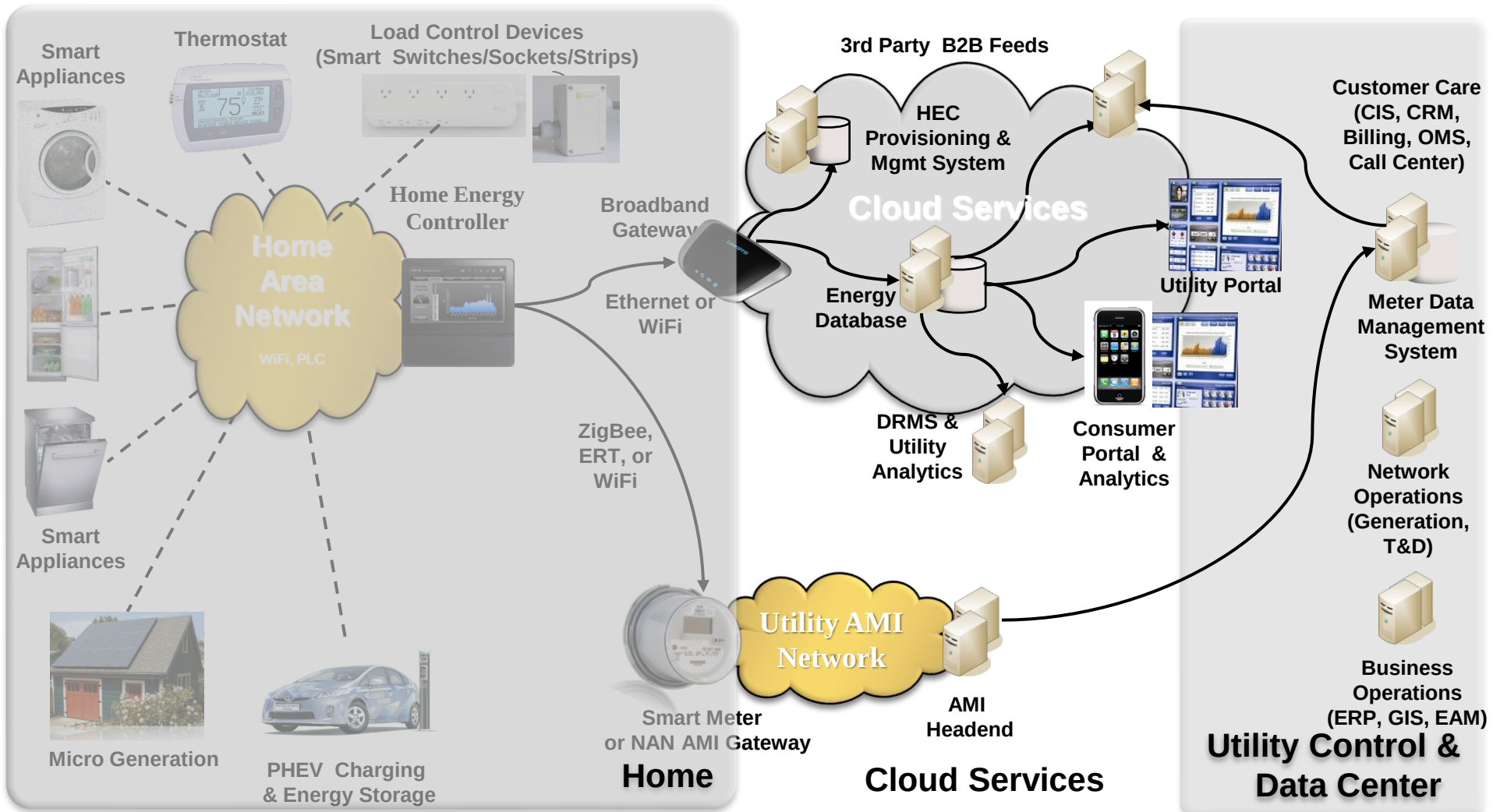
Smart Connect platform

- ▶ German telecoms giant Deutsche Telekom (DT) launches a connected home platform, called Smart Connect



The platform, based around a home hub that connects to the cloud via any broadband internet connection (DT or non-DT), is aimed at connecting all kinds of devices in the home, including HVAC/comfort, security, lighting and white goods.

Energy Management - Architecture



What is Smart Appliance?

- ▶ To allow customers and utilities to manage loads through advanced communication and controls at the end-use device level
- ▶ To allow customers and other parties to remotely monitor, control loads and diagnose faults



How Smart Appliances Work?

- ▶ Smart appliances are home and light commercial appliances equipped with commercially available sensors and computing capability to learn and adapt to the owner's lifestyle habits and to regulate their energy consumption accordingly
- ▶ Two categories
 1. The first group features appliances which can be linked into other home automation network systems and can be manipulated remotely through mobile devices or personal computers
 2. The second group has a greater focus on configuring technology to accept price signals from the utility during peak hours or to recognize grid stability and then to limit energy consumption during these situations

Smart Appliances Pros & Cons

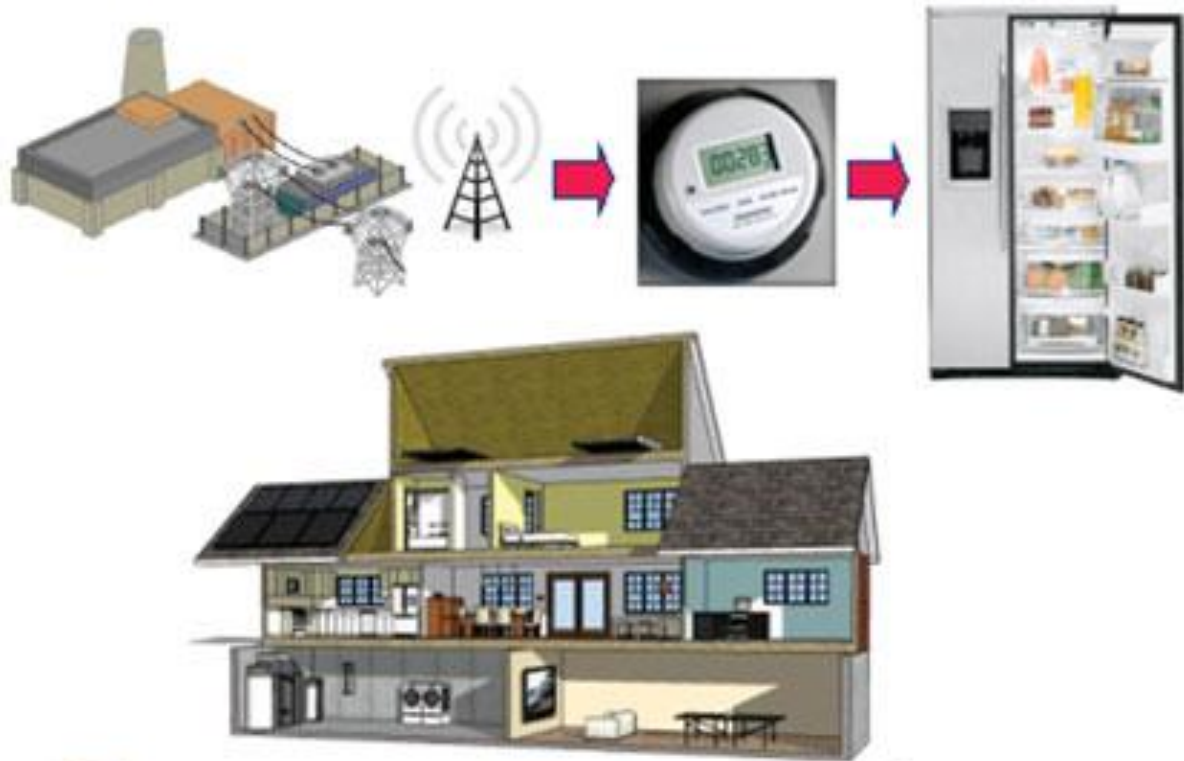
Advantages

- ▶ Offer means to manage electricity demand without generation or delivery system upgrades
- ▶ Can lower energy bills and increase convenience for residential customers
- ▶ Can increase productivity and reliability of the distribution system if matched with DR
- ▶ Fault detection diagnostics reduces downtime for small business and consumers

Disadvantages

- ▶ Potentially high upfront cost for smart appliances marketed as premium products
- ▶ May require periodic review by end users to ensure effectiveness

Smart Appliance & AMI



imagination at work

GE – Smart appliance & Energy management for a Smart Home

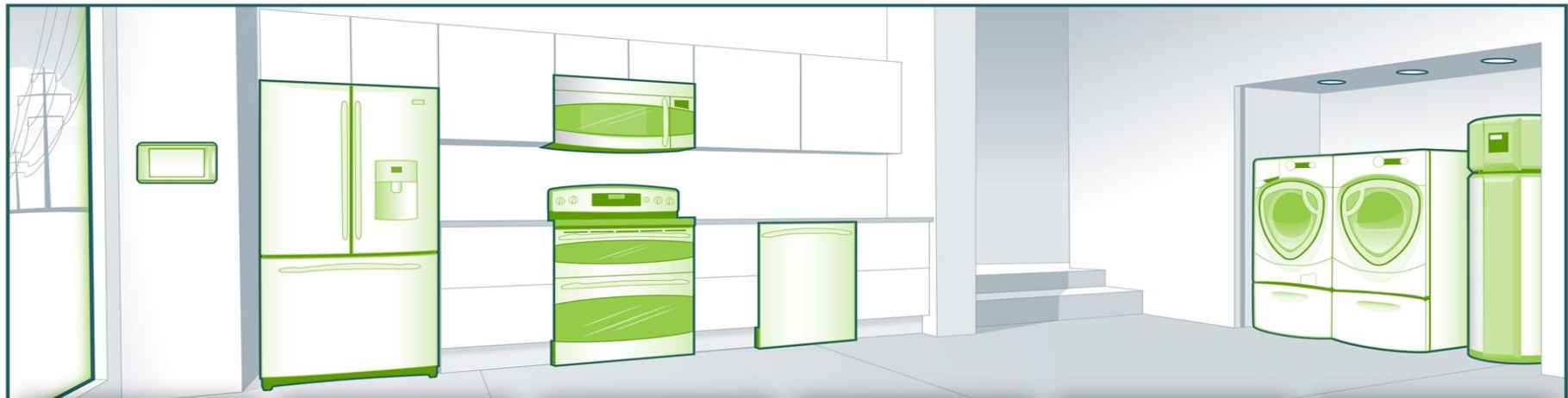
GE Energy Management & Demand Response Appliances



Utility companies that have implemented time-of-use pricing communicate their daily rate schedules to the home.

GE's Demand Response Appliances react, saving money while lowering peak demand and the need for more power generation.

Optionally, the GE Home Energy Manager correlates rates with user-preferences to balance cost, comfort, and convenience.



Home Energy Manager This energy brain works in conjunction with you and your home to help understand and optimize energy use.	Refrigerator This Profile refrigerator can reduce energy use on demand and delay defrosts to inexpensive rate periods.	Range & Microwave Cooking energy is reduced and with dual cavities, the range can preference the smaller upper oven.	Dishwasher The dishwasher is aware of expensive rate periods and can wait to run when energy costs are lower.	Laundry Pair When high rates arise, this laundry pair will save you money while getting your clothes clean and dry.	Hybrid Water Heater The water heater can switch to heat-pump mode and modify temperature settings during high rates.
					

How do they work?



HOW DO SMART HOME ENERGY MANAGEMENT SYSTEMS SAVE ENERGY?

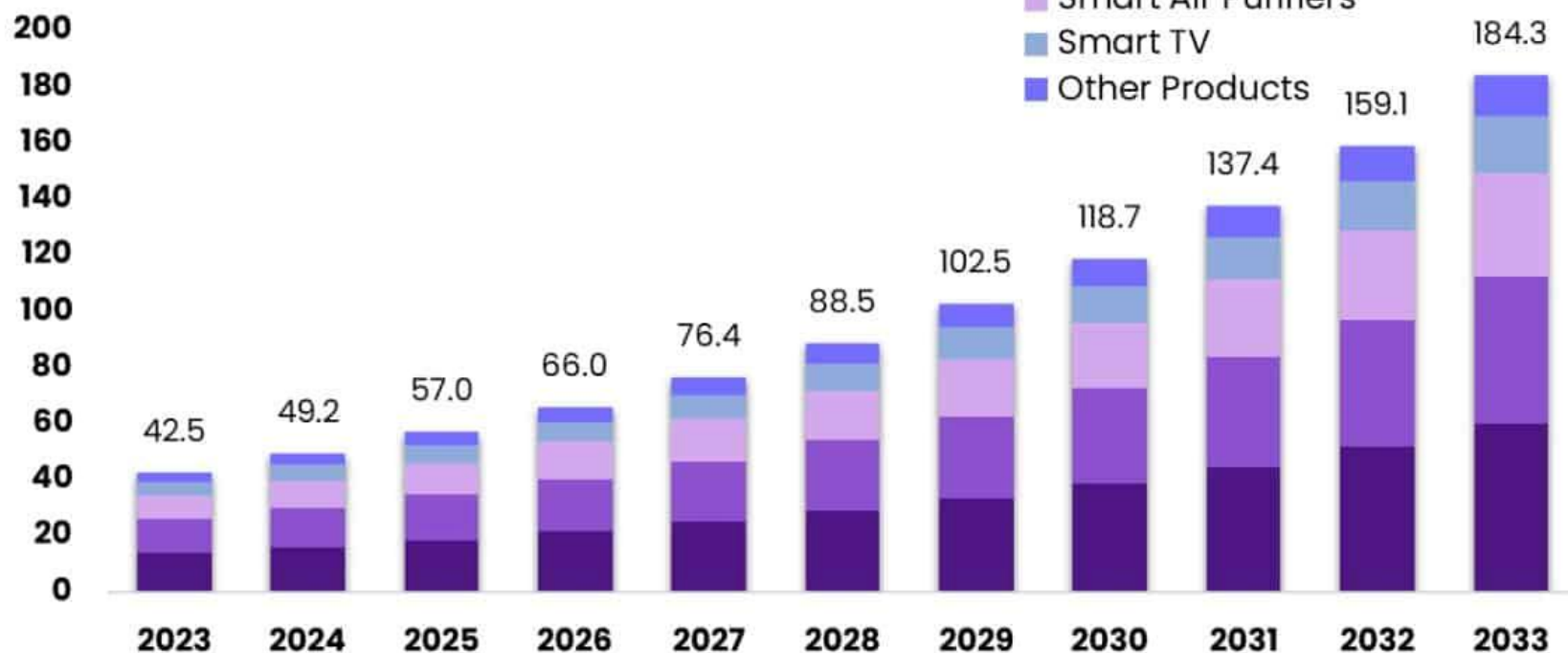


The image features a woman in a blue shirt holding a tablet. In the background, there is a house and a smart home interface. The text 'HOW DO SMART HOME ENERGY MANAGEMENT SYSTEMS SAVE ENERGY?' is displayed in a large, bold, white font with a black outline. The ECO HOME BLUEPRINT logo is visible in the bottom left corner.

Smart Appliances Market

Global Smart Appliances Market

Size, Based on Product, 2023-2033 (USD Billion)



The Market will Grow
At the CAGR of:

15.8%

The Forecasted Market
Size for 2033 in USD:

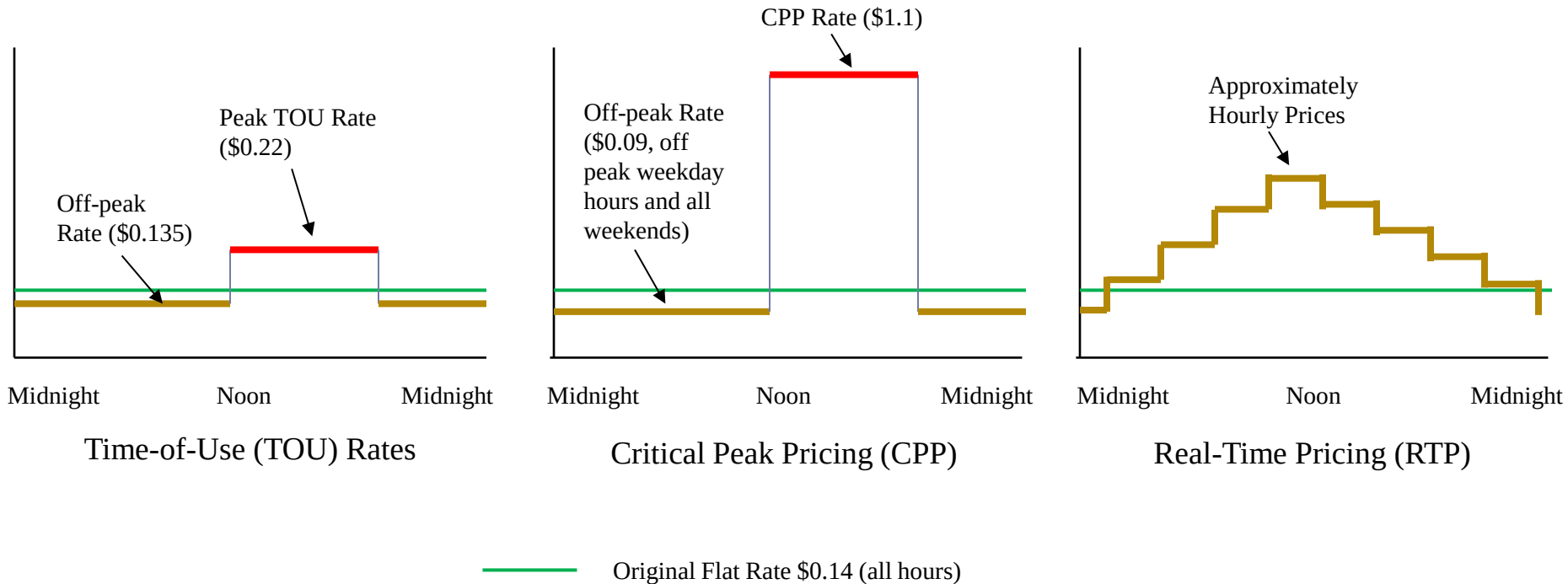
\$184.3B



market.us
ONE STOP SHOP FOR THE REPORTS

Electricity Pricing and Benefits of DR

Time Varying (Dynamic) Electricity Pricing



Effects of Dynamic Pricing

- ▶ Typical TOU rates reduce peak power demand by about 5%
- ▶ CPP achieves 10-20%
- ▶ Or, customer stop using 1 out of every 10 MW of power in peak periods
- ▶ If this could be sustained nationally (use US as an example), we could defer building about 100,000 MW of power generators – about 200 medium-sized plants.

Effects of Dynamic Pricing

- ▶ In a recent pricing pilot by Baltimore Gas and Electric, customers who adjusted their use saved 26~37% of their on-peak power and saved over \$100 a year.

Benefits of DR to customers

- ▶ It only takes a small percentage of customers shifting their use, say 5%, to reduce electricity prices substantially for everyone use.
- ▶ A study shows that:
 - ▶ Some customers (in the DR program) shift just 3% of their demands during the 100 highest-use hours in PJM market, it would save the shifting customers around \$17.5 million immediately.
 - ▶ At the same time, customers who did not shift their loads also benefited from the price reduction by the shift, saving around \$120 million (over 5 times).

Barriers and Resistance of Dynamic Pricing



AMI, SM are expensive to install



Customer concerns



People are used to “flat rate”

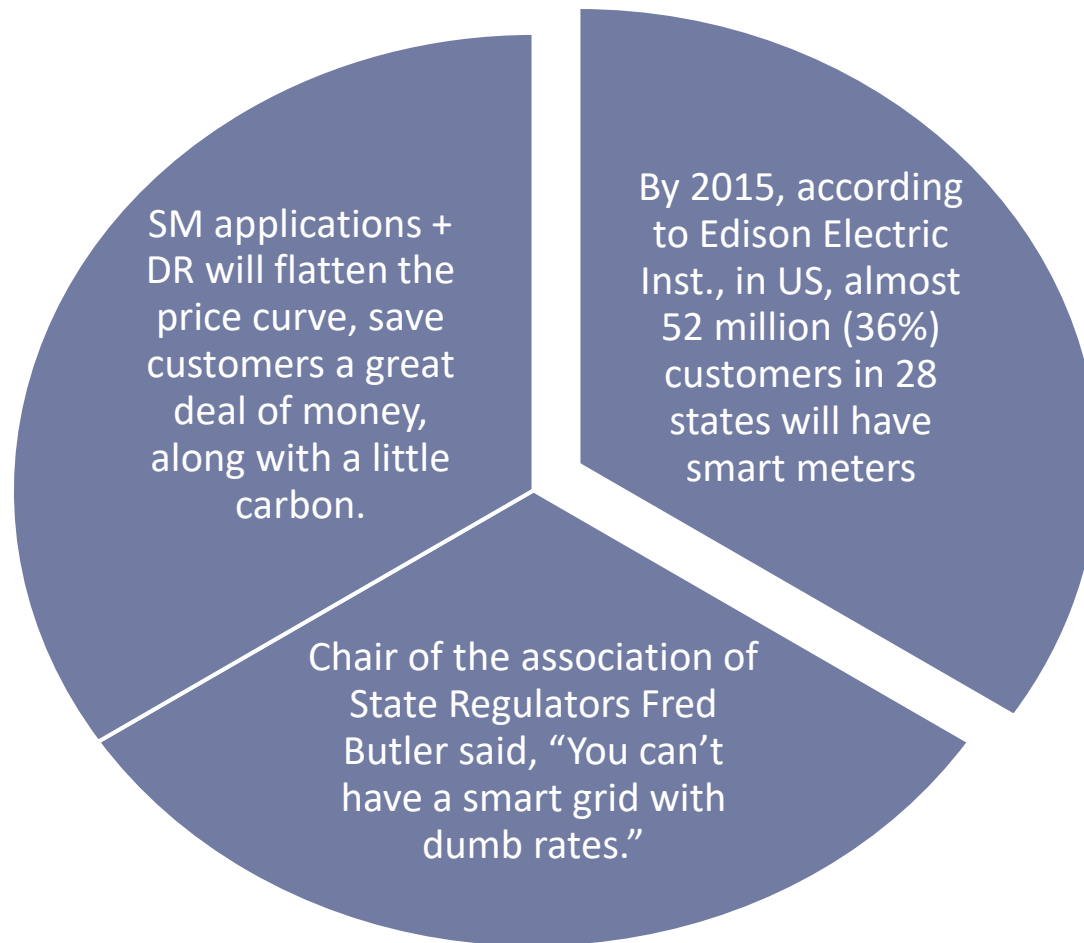


With lower price, it doesn't mean lower energy consumption, probably a little bit more consumption



Effect on carbon emission is small, but it is possible to reduce energy cost

Smart Meter and Demand Response



Demand Response (DR) in Smart Grid

- ▶ Demand Response enables consumers to modify their electricity usage in response to price signals, helping reduce peak demand and improve grid reliability. While DR has traditionally operated in industrial and commercial sectors, Smart Grid technology now extends these capabilities to residential markets through bidirectional, intelligent communication that delivers real-time pricing information to consumers.
- ▶ By participating in DR, consumers can help balance power supply with demand regardless of system size. Benefits include reduced electricity bills, improved grid reliability, greater consumer choice, enhanced market performance, and increased system security. DR represents a fundamental shift from one-way utility-consumer relationships to interactive power management across all customer segments.